

CHAPTER 2

POSITION ANALYSIS OF SERIAL MANIPULATORS

2.1 INTRODUCTION

A serial manipulator consists of several links connected in series by various types of joints, typically revolute and prismatic joints. One end of the manipulator is attached to the ground and the other end is free to move in space. For this reason a serial manipulator is sometimes called an *open-loop manipulator*. We call the fixed link the *base*, and the free end where a gripper or a mechanical hand is attached, the *end effector*.

For a robot to perform a specific task, the location of the end effector relative to the base should be established first. This is called the *position analysis problem*. There are two types of position analysis problems: *direct position* or *direct kinematics* and *inverse position* or *inverse kinematics problems*. For direct kinematics, the joint variables are given and the problem is to find the location of the end effector. For inverse kinematics, the location of the end effector is given and the problem is to find the joint variables necessary to bring the end effector to the desired location. For a serial manipulator, direct kinematics is fairly straightforward, whereas inverse kinematics becomes very difficult. On the other hand, for a parallel manipulator, inverse kinematics is very straightforward, whereas direct kinematics becomes very difficult. We note that for a deficient manipulator the end effector cannot be positioned freely in space, and for a redundant manipulator there may be several infinitudes of inverse kinematic solutions corresponding to a given end-effector location, depending on the degrees of redundancy.

In solving the inverse kinematics problem, we are often interested in obtaining a *closed-form solution*, that is, in reducing the problem to an algebraic equation relating the end-effector location to a single joint variable. In this way, all possible solutions and manipulator postures can be accounted for. To achieve this goal, various methods of formulation have been proposed. Perhaps the most commonly used methods are the *vector algebra method* (Chase, 1963; Lee and Liang, 1988a,b), the *geometric method* (Duffy and Rooney, 1975; Duffy, 1980), the 4×4 *matrix method* (Denavit and Hartenberg, 1955), the 3×3 *dual matrix method* (Yang, 1969; Pennock and Yang, 1985), the *iterative method* (Uicker et al., 1964, Albala and Angeles, 1979), the *screw algebra method* (Yuan and Freudenstein, 1971; Kohli and Soni, 1975), and the *quaternion algebra method* (Yang and Freudenstein, 1964).

The number of possible inverse kinematics solutions depends on the type and location of a robot manipulator. In general, closed-form solutions can be found for manipulators with simple geometry, such as manipulators with three consecutive joint axes intersecting at a common point or three consecutive joint axes parallel to one another. For a manipulator of general geometry, the inverse kinematics problem becomes a very difficult task. Following is a brief review of the historical development of the inverse kinematics problem.

- Pieper and Roth (1969) applied the 4×4 matrix method to solve the inverse kinematics of serial manipulators. They found that a sufficient condition for a serial manipulator to yield a closed-form inverse kinematics solution is to have any three consecutive joint axes intersecting at a common point or any three consecutive joint axes parallel to each other. They also pointed out that the analysis of a serial manipulator is related to the displacement analysis of a single-loop spatial mechanism. Therefore, the results using all methods of solution developed for single-loop spatial mechanisms can be applied to the analysis of serial manipulators. Closed-form solutions were obtained for several robot manipulators of special geometry (Pieper, 1968).
- Roth, Rastegar, and Scheinman (1972) concluded by deductive reasoning that there are at most 32 inverse kinematics solutions for the general $6R$ manipulator.
- Freudenstein (1973) referred to the inverse kinematics problem of the most general $6R$ manipulator as the “Mount Everest” of kinematic problems.
- Duffy and Crane (1980) derived a closed-form solution for the general $7R$ single-loop spatial mechanism. The solution was obtained in the form of a 16×16 determinant in which every element is a second-degree polynomial in one joint variable. The determinant, when expended,

should yield a 32nd-degree polynomial equation and hence confirms the upper limit predicted by Roth et al. (1973).

- Albala (1982) used the indicial notation to formulate the general $7R$ spatial mechanism problem and obtained the solution in a 12×12 determinant form in which every element is a quadratic polynomial in the tangent of one-half of a joint angle.
- Tsai and Morgan (1985) used the homotopy continuation method to solve the inverse kinematics of the general $6R$ manipulator and found only 16 solutions. Since the continuation method is capable of finding all possible solutions of a set of polynomial equations, they conjectured that the most general $6R$ manipulator has at most 16 real solutions.
- Primrose (1986) proved that the 32nd-degree polynomial derived by Duffy and Crane contains 16 extraneous solutions.
- Lee and Liang (1988a,b) derived a 16th-degree polynomial for the input–output displacement equation of the general $7R$ spatial mechanism.
- Raghavan and Roth (1989, 1990) used the dyalytic elimination method to derive a 16th-degree polynomial for the general $6R$ inverse kinematics problem. See Appendix C for the derivation.

In this chapter we focus on the position problem of serial manipulators. Two commonly used methods, Denavit and Hartenberg's method and the method of successive screw displacements, are introduced. Both methods are systematic in nature and more suitable for the kinematic analysis of serial manipulators. The kinematics of several frequently used robotic structures will be analyzed to demonstrate the methodologies. The inverse kinematics of the $6R$ manipulator of general geometry, which is the most difficult problem in kinematics, is presented in Appendix C. Although the geometric method is often employed by some researchers and engineers, it is judged to be more suitable for serial manipulators with relatively simple geometry and for the analysis of parallel manipulators. We describe this method only briefly in this chapter.

2.2 LINK PARAMETERS AND LINK COORDINATE SYSTEMS

In general, an n -dof serial manipulator consists of a base link and n moving links connected in series by n joints without forming a closed loop. The relative motion associated with each joint can be controlled by an actuator such that the end effector can be positioned anywhere within its workspace. For example, the Scorbot robot shown in Fig. 2.1 is a 5-dof serial manipu-

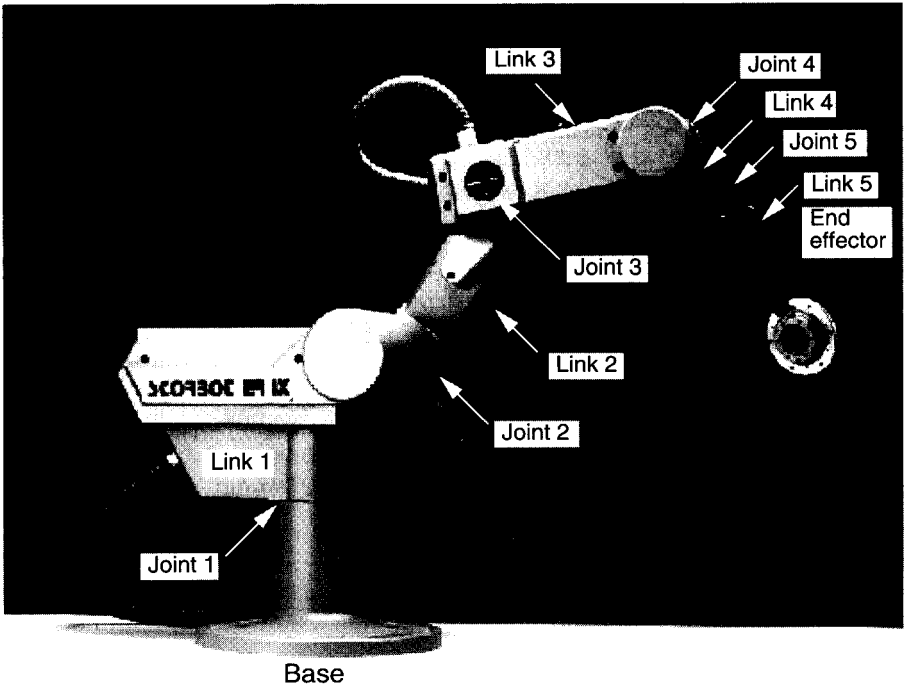


FIGURE 2.1. Scorbot robot. (Courtesy of Eshed Robotec Inc., Princeton, New Jersey.)

lator in which the first joint axis points up vertically, the second joint axis is perpendicular to the first with a small offset, the third and fourth joint axes are both parallel to the second, and the fifth joint axis intersects the fourth perpendicularly.

To describe the geometry of the links, starting from the base link, we number the links sequentially from 0 to n and the joints from 1 to n . Thus, except for the base link and the end-effector link, every link has two joints. Link 1 is connected to the base link by joint 1, link 2 is connected to link 1 by joint 2, and so on. Link i has joint i at its proximal end and joint $i + 1$ at the distal end, as shown in Fig. 2.2.

Following Denavit and Hartenberg's convention (1955), a Cartesian coordinate system is attached to each link of a manipulator. Except for the base and end-effector link, coordinate system i is attached to link i according to the following rules:

- The z_i -axis is aligned with the $(i + 1)$ th joint axis. The positive direction of rotation or translation can be chosen arbitrarily.

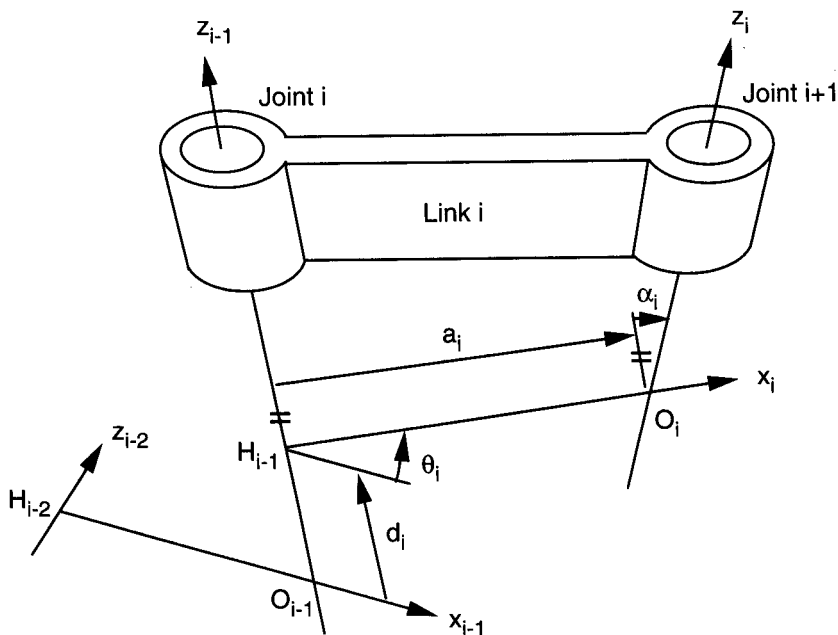


FIGURE 2.2. Definition of link parameters.

- The x_i -axis is defined along the common normal between the i th and $(i + 1)$ th joint axes and points from the i th to the $(i + 1)$ th joint axis. If the two joint axes are parallel, the x_i -axis can be chosen anywhere perpendicular to the two joint axes. In case of two intersecting joint axes, the x_i -axis can be defined either in the direction of the vector cross product $\mathbf{z}_{i-1} \times \mathbf{z}_i$ or in the opposite direction, and the origin is at the point of intersection.
- The y_i -axis is determined by the right-hand rule.

The zeroth coordinate system is attached to the base at any convenient location as long as the z_0 -axis is aligned with the first joint axis. Furthermore, we attach a coordinate system to the end-effector link, called the *end effector* or *hand coordinate system*, to describe the location of the end effector. The hand coordinate system can be located anywhere in the end-effector as long as the x_n -axis is normal to the last joint axis. For convenience, the z_n -axis is often defined along the direction of approach of a gripper.

Let H_{i-1} be the point of intersection of the x_i and z_{i-1} axes, and let O_i , the origin of the i th coordinate system, be the point of intersection of the x_i and z_i axes as shown in Fig. 2.2. Then, regardless of how the links are constructed physically, the following parameters are uniquely determined by the geometry of the axes:

- a_i : offset distance between two adjacent joint axes, where $a_i = |H_{i-1} O_i|$.
- d_i : translational distance between two incident normals of a joint axis. $d_i = |O_{i-1} H_{i-1}|$ is positive if the vector $\overline{O_{i-1} H_{i-1}}$ points in the positive z_{i-1} -direction; otherwise, it is negative.
- α_i : twist angle between two adjacent joint axes. It is the angle required to rotate the z_{i-1} -axis into alignment with the z_i -axis about the positive x_i -axis according to the right-hand rule.
- θ_i : joint angle between two incident normals of a joint axis. It is the angle required to rotate the x_{i-1} -axis into alignment with the x_i -axis about the positive z_{i-1} -axis according to the right-hand rule.

For a revolute joint, a_i , α_i , and d_i are constant, and θ_i is a variable that measures the relative location of link i with respect to link $i - 1$. For a prismatic joint, a_i , α_i , and θ_i are constant, and d_i is a variable that measures the relative location of link i with respect to link $i - 1$. In what follows, we refer to θ_i for a revolute joint and d_i for a prismatic joint as *joint variables* and to the constant parameters as *link parameters*. We note that a joint variable provides only the relative location between two adjacent links, it should not be confused with the term *displacement*. A displacement implies the amount of angle or distance needed to move a link from one location to another. Using the definitions above, the displacement required to move a link from one location to another is equal to the difference between two successive values of a joint variable.

For a prismatic joint, the direction of the joint axis defines the direction of relative translation between two links. Unlike a revolute joint, only the direction of the joint axis is important. Although the location of a prismatic joint has no effect on relative displacement, the physical location of the joint will be used to establish the foregoing link parameters. Note that several different coordinate systems can be defined for a manipulator, due to various possible choices of the positive z and x axes. A general procedure for establishing Denavit–Hartenberg coordinate systems can be summarized as follows:

1. Starting from the base link, number the links and joints sequentially. The base is numbered as link 0 and the last link is the end effector. Except for the base and end effector link, every link contains two joints. Joint i connects link i to link $i - 1$.
2. Draw the common normals between every two adjacent joint axes. Except for the first and last joint axes, every joint axis should have two incident common normals, one with the $(i - 1)$ th and the other with the $(i + 1)$ th joint axis.

3. Establish the base coordinate system such that the z_0 -axis is aligned with the first joint axis, the x_0 -axis is perpendicular to the z_0 -axis, and the y_0 -axis is determined by the right-hand rule.
4. Establish the n th hand coordinate system such that the x_n -axis is perpendicular to the last joint axis. The z_n -axis is usually chosen in the direction of approach of the end effector.
5. Attach a Cartesian coordinate system to the distal end of all the other links as follows:
 - The z_i -axis is aligned with the $(i + 1)$ th joint axis.
 - The x_i -axis is defined along the common normal between the i th and $(i + 1)$ th joint axes, pointing from the i th to the $(i + 1)$ th joint axis. If the joint axes are parallel, the x_i -axis can be chosen anywhere perpendicular to the two joint axes. In the case of two intersecting joint axes, the x_i -axis can be defined either in the direction of the vector cross product $\mathbf{z}_{i-1} \times \mathbf{z}_i$ or in the opposite direction, and the origin is located at the point of intersection.
 - The y_i -axis is defined according to the right-hand rule.
6. Determine the link parameters and joint variables, a_i , α_i , θ_i , and d_i .

There are $n + 1$ coordinate systems for an n -dof manipulator. However, if additional reference coordinate systems are defined, they can be related to one of the coordinate systems above by a transformation matrix. We note that John Craig used a different convention; he attached the i th coordinate system to the proximal end of link i , which results in a different homogeneous transformation matrix.

2.3 DENAVIT–HARTENBERG HOMOGENEOUS TRANSFORMATION MATRICES

Having established a coordinate system to each link of a manipulator, a 4×4 transformation matrix relating two successive coordinate systems can be established. Observation of Fig. 2.2 reveals that the i th coordinate system can be thought of as being displaced from the $(i - 1)$ th coordinate system by the following successive rotations and translations.

1. The $(i - 1)$ th coordinate system is translated along the z_{i-1} -axis a distance d_i . This brings the origin O_{i-1} into coincidence with H_{i-1} . The corresponding transformation matrix is

$$T(z, d) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

2. The displaced $(i - 1)$ th coordinate system is rotated about the z_{i-1} -axis an angle θ_i , which brings the x_{i-1} -axis into alignment with the x_i -axis. The corresponding transformation matrix is

$$T(z, \theta) = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & 0 \\ s\theta_i & c\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

3. The displaced $(i - 1)$ th coordinate system is translated along the x_i -axis a distance a_i . This brings the origin O_{i-1} into coincidence with O_i . The corresponding transformation matrix is

$$T(x, a) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4. The displaced $(i - 1)$ th coordinate system is rotated about the x_i -axis an angle α_i , which brings the two coordinate systems into complete coincidence. The corresponding transformation matrix is

$$T(x, \alpha) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\alpha_i & -s\alpha_i & 0 \\ 0 & s\alpha_i & c\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

We may think of the transformations above as four basic transformations about the moving coordinate axes. Therefore, the resulting transformation matrix, ${}^{i-1}A_i$, is given by

$${}^{i-1}A_i = T(z, d)T(z, \theta)T(x, a)T(x, \alpha). \quad (2.1)$$

Expanding Eq. (2.1), we obtain

$${}^{i-1}A_i = \begin{bmatrix} c\theta_i & -c\alpha_i s\theta_i & s\alpha_i s\theta_i & a_i c\theta_i \\ s\theta_i & c\alpha_i c\theta_i & -s\alpha_i c\theta_i & a_i s\theta_i \\ 0 & s\alpha_i & c\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.2)$$

Equation (2.2) is called the *Denavit–Hartenberg (D-H) transformation matrix*. The trailing subscript i and the leading superscript $i - 1$ denote that the transformation takes place from the i th coordinate system to the $(i - 1)$ th coordinate system.

Let the homogeneous coordinates of the position vector of a point relative to the i th coordinate system be denoted by ${}^i\mathbf{p} = [p_x, p_y, p_z, 1]^T$. Also let the homogeneous coordinates of a unit vector expressed in the i th coordinate system be denoted by ${}^i\mathbf{u} = [u_x, u_y, u_z, 0]^T$. Then the transformation of a position vector and a unit vector from the i th to the $(i - 1)$ th coordinate system can be written as

$${}^{i-1}\mathbf{p} = {}^{i-1}A_i {}^i\mathbf{p}, \quad (2.3)$$

$${}^{i-1}\mathbf{u} = {}^{i-1}A_i {}^i\mathbf{u}. \quad (2.4)$$

Note that the leading superscript is used to indicate the coordinate system with respect to which a vector is expressed. Although the transformation matrix A is not orthogonal, the inverse transformation exists and is given by

$${}^iA_{i-1} = ({}^{i-1}A_i)^{-1} = \begin{bmatrix} c\theta_i & s\theta_i & 0 & -a_i \\ -c\alpha_i s\theta_i & c\alpha_i c\theta_i & s\alpha_i & -d_i s\alpha_i \\ s\alpha_i s\theta_i & -s\alpha_i c\theta_i & c\alpha_i & -d_i c\alpha_i \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.5)$$

Example 2.3.1 Planar 3-DOF Manipulator Figure 2.3 shows a 3-dof planar manipulator constructed with three revolute joints located at points O_0 , A , and P , respectively. A coordinate system is attached to each link. The (x_0, y_0, z_0) coordinate system is attached to the base with its origin located at the first joint pivot and the x -axis pointing to the right. Since the joint axes are all parallel to each other, all the twist angles α_i and translational distances d_i are zero.

For the coordinate systems chosen, the link parameters are given in Table 2.1. The D-H transformation matrices are obtained by substituting the D-H link parameters into Eq. (2.2):

$${}^0A_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_1 c\theta_1 \\ s\theta_1 & c\theta_1 & 0 & a_1 s\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.6)$$

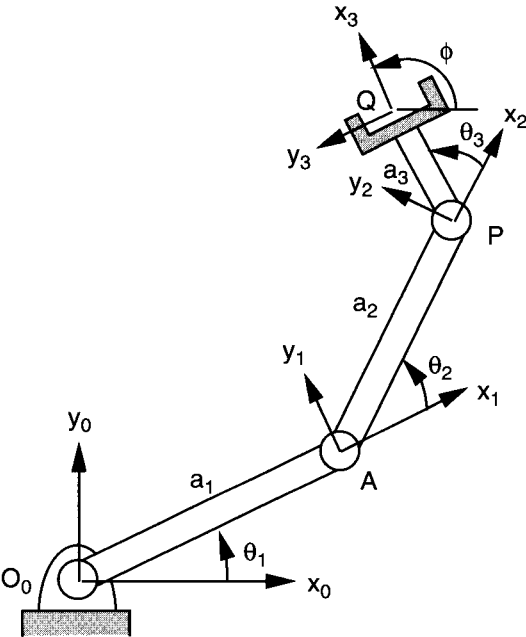


FIGURE 2.3. Planar 3-dof manipulator.

$${}^1A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{2.7}$$

$${}^2A_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_3c\theta_3 \\ s\theta_3 & c\theta_3 & 0 & a_3s\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2.8}$$

Example 2.3.2 SCARA Arm The SCARA arm is an important type of 4-dof manipulator. It has been produced by several companies, including Adept

TABLE 2.1. D-H Parameters of a 3-DOF Manipulator

Joint i	α_i	a_i	d_i	θ_i
1	0	a_1	0	θ_1
2	0	a_2	0	θ_2
3	0	a_3	0	θ_3

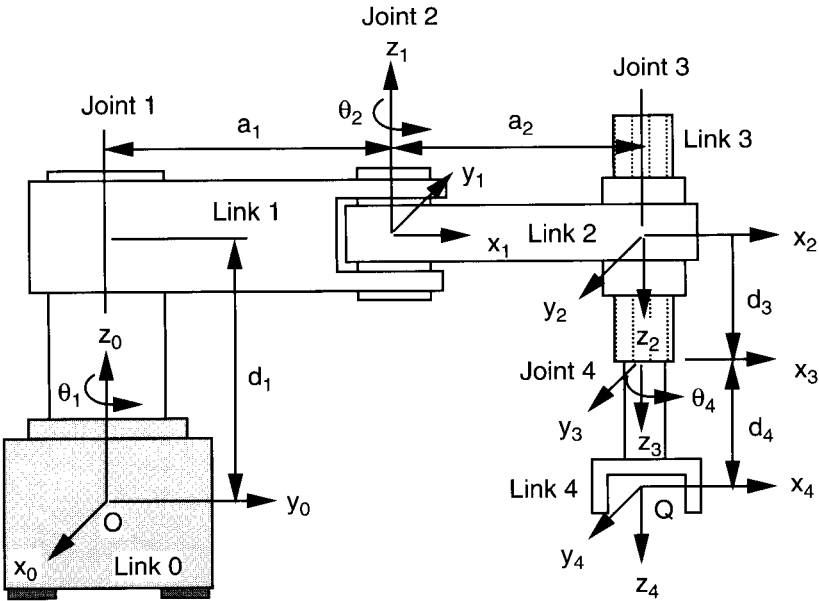


FIGURE 2.4. Schematic diagram of a SCARA arm.

Technology, IBM, Seiko, and others. A SCARA arm is constructed with four joint axes parallel to each other. The first two and the fourth are revolute joints, and the third is a prismatic joint. Figure 2.4 shows a schematic diagram of a SCARA arm. For the coordinate systems established in the figure, the corresponding link parameters are listed in Table 2.2.

Substituting the D-H link parameters into Eq. (2.2), we obtain the D-H transformation matrices:

$${}^0A_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_1c\theta_1 \\ s\theta_1 & c\theta_1 & 0 & a_1s\theta_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{2.9}$$

TABLE 2.2. D-H Parameters of the SCARA Arm

Joint i	α_i	a_i	d_i	θ_i
1	0	a_1	d_1	θ_1
2	π	a_2	0	θ_2
3	0	0	d_3	0
4	0	0	d_4	θ_4

$${}^1A_2 = \begin{bmatrix} c\theta_2 & s\theta_2 & 0 & a_2c\theta_2 \\ s\theta_2 & -c\theta_2 & 0 & a_2s\theta_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.10)$$

$${}^2A_3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.11)$$

$${}^3A_4 = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & 0 \\ s\theta_4 & c\theta_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.12)$$

In this robot, the joint variables are θ_1 , θ_2 , d_3 , and θ_4 . The first two joint variables control the x and y coordinates, the third joint variable controls the z coordinate, and the fourth joint variable controls the orientation of the end effector. Since the robot has only 4 degrees of freedom, the orientation of the end effector cannot be specified arbitrarily. As a matter of fact, the z_4 -axis must be always pointing in the negative z_0 direction. Although the SCARA robot has only 4 degrees of freedom, it is very useful for assembling components on a plane such as a PC board.

2.4 LOOP-CLOSURE EQUATIONS

In a study of the kinematics of robot manipulators, we are interested in deriving an algebraic equation relating the location of the end effector to the joint variables. The location of the end effector can be specified by the following 4×4 homogeneous transformation matrix:

$${}^0A_n = \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} & \mathbf{q} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.13)$$

where the upper right 3×1 submatrix describes the position of a reference point Q and the upper left 3×3 submatrix describes the orientation of the end effector. The orientation of the end effector can be specified in terms of three Euler angles, or the direction cosines of the three end-effector coordinate axes, u , v , and w . If the w - u - w Euler angles are used, for example, the elements of the upper left 3×3 submatrix are given by

$$u_x = c\phi c\psi - s\phi c\theta s\psi,$$

$$u_y = s\phi c\psi + c\phi c\theta s\psi,$$

$$\begin{aligned}
u_z &= s\theta s\psi, \\
v_x &= -c\phi s\psi - s\phi c\theta c\psi, \\
v_y &= -s\phi s\psi + c\phi c\theta c\psi, \\
v_z &= s\theta c\psi, \\
w_x &= s\phi s\theta, \\
w_y &= -c\phi s\theta, \\
w_z &= c\theta.
\end{aligned} \tag{2.14}$$

If the direction cosines are used, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ represent three unit vectors directed along the three coordinate axes of the hand coordinate system and expressed in the base coordinate system.

From the geometry of the links, the transformation matrix 0A_n above can be thought of as the resultant of a series of coordinate transformations beginning from the base coordinate system to the end-effector coordinate system. That is,

$${}^0A_1 {}^1A_2 {}^2A_3 \cdots {}^{n-1}A_n = {}^0A_n. \tag{2.15}$$

Equation (2.15) is called the *loop-closure equation* of a serial manipulator. It contains 16 scalar equations, four of which are trivial. Equating the upper right 3×1 submatrix results in three independent equations, representing the position of the end effector. Equating the elements of the upper left 3×3 submatrix results in nine equations, representing the orientation of the end effector. However, only three of the nine orientation equations are independent because of the orthogonal conditions.

The loop-closure equation, Eq. (2.15), can be used to solve both direct and inverse kinematics problems. For direct kinematics, the joint variables are given and the problem is to find where the end effector is with respect to the base coordinate system. This can be accomplished by multiplying the D-H matrices on the left-hand side of the equation. For the inverse kinematics, the end-effector location (i.e., 0A_n) is given and the problem is to find the joint variables needed to bring the end effector to the desired location. The problem becomes very nonlinear. In what follows, we concentrate on the inverse kinematics problem.

Example 2.4.1 *Scorbot Robot* Figure 2.5 shows a schematic diagram of the Scorbot robot. In this diagram, the second, third, and fourth joint axes are parallel to one another and point into the paper at points A , B , and P , respectively. The first joint axis points up vertically, and the fifth joint axis intersects the fourth perpendicularly. We wish to find the overall transformation matrix for the robot.

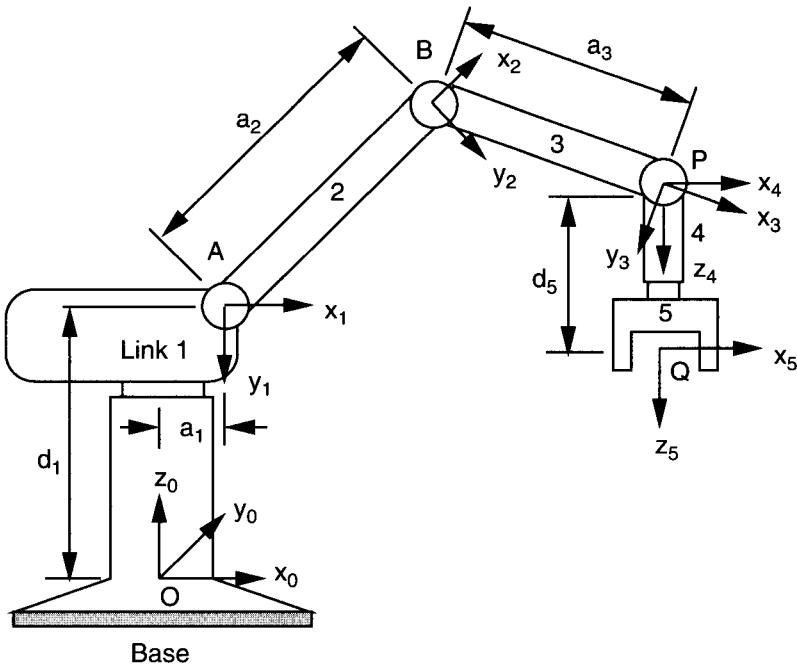


FIGURE 2.5. Schematic diagram of the Scorbot robot.

Using the coordinate systems established in Fig. 2.5, the corresponding link parameters are listed in Table 2.3. Substituting the D-H link parameters into Eq. (2.2), we obtain the D-H transformation matrices:

$${}^0A_1 = \begin{bmatrix} c\theta_1 & 0 & -s\theta_1 & a_1c\theta_1 \\ s\theta_1 & 0 & c\theta_1 & a_1s\theta_1 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.16)$$

TABLE 2.3. D-H Parameters of a 5-DOF Manipulator

Joint i	α_i	a_i	d_i	θ_i
1	$-\pi/2$	a_1	d_1	θ_1
2	0	a_2	0	θ_2
3	0	a_3	0	θ_3
4	$-\pi/2$	0	0	θ_4
5	0	0	d_5	θ_5

$${}^1A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.17)$$

$${}^2A_3 = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_3c\theta_3 \\ s\theta_3 & c\theta_3 & 0 & a_3s\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.18)$$

$${}^3A_4 = \begin{bmatrix} c\theta_4 & 0 & -s\theta_4 & 0 \\ s\theta_4 & 0 & c\theta_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.19)$$

$${}^4A_5 = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.20)$$

Multiplying Eqs. (2.17), (2.18), and (2.19) yields

$${}^1A_4 = \begin{bmatrix} c\theta_{234} & 0 & -s\theta_{234} & a_3c\theta_{23} + a_2c\theta_2 \\ s\theta_{234} & 0 & c\theta_{234} & a_3s\theta_{23} + a_2s\theta_2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.21)$$

where $c\theta_{ij} = \cos(\theta_i + \theta_j)$, $s\theta_{ij} = \sin(\theta_i + \theta_j)$, $c\theta_{ijk} = \cos(\theta_i + \theta_j + \theta_k)$, and $s\theta_{ijk} = \sin(\theta_i + \theta_j + \theta_k)$.

Note that Eq. (2.21) provides a transformation from the fourth coordinate system to the first coordinate system. We may treat θ_2 , θ_{23} , and θ_{234} as new variables. In this way, the orientation submatrix contains only one variable, θ_{234} , while the position submatrix contains two variables, θ_2 and θ_{23} . This important fact can be used for deriving a closed-form solution for any manipulator with three consecutive parallel joint axes.

Multiplying Eqs. (2.16), (2.21), and (2.20) yields the elements of the overall transformation matrix 0A_5 :

$$u_x = c\theta_1c\theta_{234}c\theta_5 + s\theta_1s\theta_5,$$

$$u_y = s\theta_1c\theta_{234}c\theta_5 - c\theta_1s\theta_5,$$

$$u_z = -s\theta_{234}c\theta_5,$$

$$v_x = -c\theta_1c\theta_{234}s\theta_5 + s\theta_1c\theta_5,$$

$$\begin{aligned}
v_y &= -s\theta_1 c\theta_{234} s\theta_5 - c\theta_1 c\theta_5, \\
v_z &= s\theta_{234} s\theta_5, \\
w_x &= -c\theta_1 s\theta_{234}, \\
w_y &= -s\theta_1 s\theta_{234}, \\
w_z &= -c\theta_{234}, \\
q_x &= c\theta_1 (a_1 + a_2 c\theta_2 + a_3 c\theta_{23} - d_5 s\theta_{234}), \\
q_y &= s\theta_1 (a_1 + a_2 c\theta_2 + a_3 c\theta_{23} - d_5 s\theta_{234}), \\
q_z &= d_1 - a_2 s\theta_2 - a_3 s\theta_{23} - d_5 c\theta_{234}.
\end{aligned} \tag{2.22}$$

Since this is a 5-dof manipulator, only five of the six parameters of the end effector can be specified. Very often, the desired position of a point and the direction of a line in the end effector (e.g., the position of point Q and the direction of x_5 -axis) are specified. Five-dof manipulators are useful for spray painting, spot welding, and sealant applications for which only the position and direction of a line are essential.

2.5 OTHER COORDINATE SYSTEMS

In the preceding section, the z_0 -axis of the base coordinate system was chosen to be in line with the first joint axis, and the z_n -axis of the hand coordinate system was chosen to be in the direction of approach. If an additional coordinate system is defined in the base with a transformation matrix ${}^{\text{ref}}A_0$, and another coordinate system is defined in the tool frame with a transformation matrix ${}^nA_{\text{tool}}$, the overall loop-closure equation can be modified as

$${}^{\text{ref}}A_{\text{tool}} = {}^{\text{ref}}A_0 {}^0A_n {}^nA_{\text{tool}}, \tag{2.23}$$

where ${}^{\text{ref}}A_0$ and ${}^nA_{\text{tool}}$ are constant transformation matrices.

2.6 DENAVIT-HARTENBERG METHOD

Although the loop-closure equation, Eq. (2.15), can be applied to solve the inverse kinematics problem, in practice it is rarely solved in its present form. In general, if there are three intersecting joint axes, we may work with the position of the point of intersection first, thereby avoiding the joint variables associated with the three intersecting axes. If there are three parallel joint axes, we may combine the three joint variables as illustrated in the Scorbot

robot example. We may also pre- or postmultiply the loop-closure equation by the inverse of the matrix ${}^{i-1}A_i$ to obtain alternative loop-closure equations, such as

$$({}^0A_1)^{-1} {}^0A_n = {}^1A_2 {}^2A_3 \cdots {}^{n-1}A_n, \quad (2.24)$$

$$({}^1A_2)^{-1} ({}^0A_1)^{-1} {}^0A_n = {}^2A_3 {}^3A_4 \cdots {}^{n-1}A_n, \quad (2.25)$$

$$({}^2A_3)^{-1} ({}^1A_2)^{-1} ({}^0A_1)^{-1} {}^0A_n = {}^3A_4 {}^4A_5 \cdots {}^{n-1}A_n. \quad (2.26)$$

One reason for rearranging the loop-closure equation is to redistribute the unknown variables on both sides of the equation as evenly as possible. Another reason is to take advantage of some special conditions, such as three consecutive intersecting joint axes or three consecutive parallel joint axes. In many cases, the equation becomes decoupled and a closed-form solution can be derived.

2.6.1 Position Analysis of a Planar 3-DOF Manipulator

For the planar 3-dof manipulator shown in Fig. 2.3, the overall transformation matrix is given by

$${}^0A_3 = {}^0A_1 {}^1A_2 {}^2A_3. \quad (2.27)$$

Substituting Eqs. (2.6) through (2.8) into (2.27), we obtain

$${}^0A_3 = \begin{bmatrix} c\theta_{123} & -s\theta_{123} & 0 & a_1c\theta_1 + a_2c\theta_{12} + a_3c\theta_{123} \\ s\theta_{123} & c\theta_{123} & 0 & a_1s\theta_1 + a_2s\theta_{12} + a_3s\theta_{123} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.28)$$

(a) Direct Kinematics. The position vector of the origin Q expressed in the end-effector coordinate system is given by ${}^3\mathbf{q} = [0, 0, 0, 1]^T$. Let the position vector of Q with respect to the base coordinate system be ${}^0\mathbf{q} = [q_x, q_y, q_z, 1]^T$. Then we can relate ${}^3\mathbf{q}$ to ${}^0\mathbf{q}$ by the following transformation:

$$\begin{bmatrix} q_x \\ q_y \\ q_z \\ 1 \end{bmatrix} = {}^0A_3 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a_1c\theta_1 + a_2c\theta_{12} + a_3c\theta_{123} \\ a_1s\theta_1 + a_2s\theta_{12} + a_3s\theta_{123} \\ 0 \\ 1 \end{bmatrix}. \quad (2.29)$$

Hence, given θ_1, θ_2 , and θ_3 , the position of point Q can be computed by Eq. (2.29). Similarly, the position vector of any other point in the end effector,

${}^3\mathbf{g} = [g_u, g_v, 0, 1]^T$, is given by

$$\begin{bmatrix} g_x \\ g_y \\ g_z \\ 1 \end{bmatrix} = {}^0A_3 \begin{bmatrix} g_u \\ g_v \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} g_u c\theta_{123} - g_v s\theta_{123} + a_1 c\theta_1 + a_2 c\theta_{12} + a_3 c\theta_{123} \\ g_u s\theta_{123} + g_v c\theta_{123} + a_1 s\theta_1 + a_2 s\theta_{12} + a_3 s\theta_{123} \\ 0 \\ 1 \end{bmatrix}. \quad (2.30)$$

From Eq. (2.28), we conclude that the orientation angle of the end effector is equal to $\theta_1 + \theta_2 + \theta_3$.

(b) Inverse Kinematics. For the inverse kinematics problem, the location of the end effector is given and the problem is to find the joint angles θ_i , $i = 1, 2, 3$, necessary to bring the end effector to the desired location. For a planar 3-dof manipulator, the end effector can be specified in terms of the position of point Q and an orientation angle ϕ of the end effector. Hence the overall transformation matrix from the end-effector coordinate system to the base coordinate system, 0A_3 , is given by

$${}^0A_3 = \begin{bmatrix} c\phi & -s\phi & 0 & q_x \\ s\phi & c\phi & 0 & q_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.31)$$

Inverse kinematics solutions can be obtained by equating the elements of Eq. (2.28) to that of (2.31). To find the orientation of the end effector, we equate the (1,1) and (2,1) elements of Eq. (2.28) to that of (2.31):

$$c\theta_{123} = c\phi, \quad (2.32)$$

$$s\theta_{123} = s\phi. \quad (2.33)$$

Hence

$$\theta_{123} = \theta_1 + \theta_2 + \theta_3 = \phi. \quad (2.34)$$

Next, we equate the (1,4) and (2,4) elements of Eq. (2.28) to that of (2.31):

$$p_x = a_1 c\theta_1 + a_2 c\theta_{12}, \quad (2.35)$$

$$p_y = a_1 s\theta_1 + a_2 s\theta_{12}, \quad (2.36)$$

where $p_x = q_x - a_3 c\phi$ and $p_y = q_y - a_3 s\phi$ denote the position vector of the point P located at the third joint axis shown in Fig. 2.3. Note that by using

this substitution, θ_3 disappears from Eqs. (2.35) and (2.36). From Fig. 2.3 we observe that the distance from point O to P is independent of θ_1 . Hence we can eliminate θ_1 by summing the squares of Eqs. (2.35) and (2.36); that is,

$$p_x^2 + p_y^2 = a_1^2 + a_2^2 + 2a_1a_2c\theta_2. \quad (2.37)$$

Solving Eq. (2.37) for θ_2 , we obtain

$$\theta_2 = \cos^{-1}\kappa, \quad (2.38)$$

where

$$\kappa = \frac{p_x^2 + p_y^2 - a_1^2 - a_2^2}{2a_1a_2}.$$

Equation (2.38) yields (1) two real roots if $|\kappa| < 1$, (2) one double root if $|\kappa| = 1$, and (3) no real roots if $|\kappa| > 1$. In general, if $\theta_2 = \theta_2^*$ is a solution, $\theta_2 = -\theta_2^*$ is also a solution, where $\pi \geq \theta_2^* \geq 0$. We call $\theta_2 = \theta_2^*$ the elbow-down solution and $\theta_2 = -\theta_2^*$ the elbow-up solution. If $|\kappa| = 1$, the arm is in a fully stretched or folded configuration. If $|\kappa| > 1$, the position is not reachable.

Corresponding to each θ_2 , we can solve θ_1 by expanding Eqs. (2.35) and (2.36) as follows:

$$(a_1 + a_2c\theta_2)c\theta_1 - (a_2s\theta_2)s\theta_1 = p_x, \quad (2.39)$$

$$(a_2s\theta_2)c\theta_1 + (a_1 + a_2c\theta_2)s\theta_1 = p_y. \quad (2.40)$$

Solving Eqs. (2.39) and (2.40) for $c\theta_1$ and $s\theta_1$, yields

$$c\theta_1 = \frac{p_x(a_1 + a_2c\theta_2) + p_ya_2s\theta_2}{\Delta},$$

$$s\theta_1 = \frac{-p_xa_2s\theta_2 + p_y(a_1 + a_2c\theta_2)}{\Delta},$$

where $\Delta = a_1^2 + a_2^2 + 2a_1a_2c\theta_2$. Hence, corresponding to each θ_2 , we obtain a unique solution for θ_1 :

$$\theta_1 = \text{Atan2}(s\theta_1, c\theta_1). \quad (2.41)$$

In a computer program we may use the function $\text{Atan2}(x, y)$ to obtain a unique solution for θ_1 . However, the solution may be real or complex. A complex solution corresponds to an end-effector location that is not reachable by the manipulator. Once θ_1 and θ_2 are known, Eq. (2.34) yields a unique solution for θ_3 . Hence, corresponding to a given end-effector location, there are

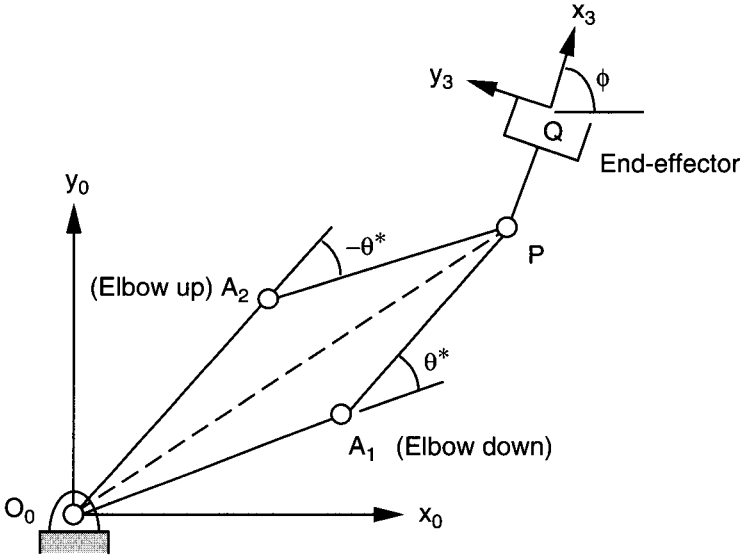


FIGURE 2.6. Two possible inverse kinematics solutions.

generally two real inverse kinematics solutions, one being the reflection of the other about a line connecting points O and P , as illustrated in Fig. 2.6.

Vector-Loop Method. Although the D-H method of analysis is a very powerful tool, the inverse kinematics problem can often be solved by other methods, such as the vector-loop method. For example, the vector-loop method becomes more efficient for analysis of the 3-dof planar manipulator shown in Fig. 2.3. For convenience, the manipulator has been resketched as shown in Fig. 2.7.

Using vector algebra, the position vector of the wrist center P can be related to the origin Q of the end effector by the equations

$$p_x = q_x - a_3 c\phi, \quad (2.42)$$

$$p_y = q_y - a_3 s\phi. \quad (2.43)$$

From Fig. 2.7, we observe that the orientation angle ϕ is related to the joint angles by

$$\phi = \theta_1 + \theta_2 + \theta_3. \quad (2.44)$$

We now form a fictitious vector loop equation as follows:

$$\overline{OA} + \overline{AP} = \overline{OP}. \quad (2.45)$$

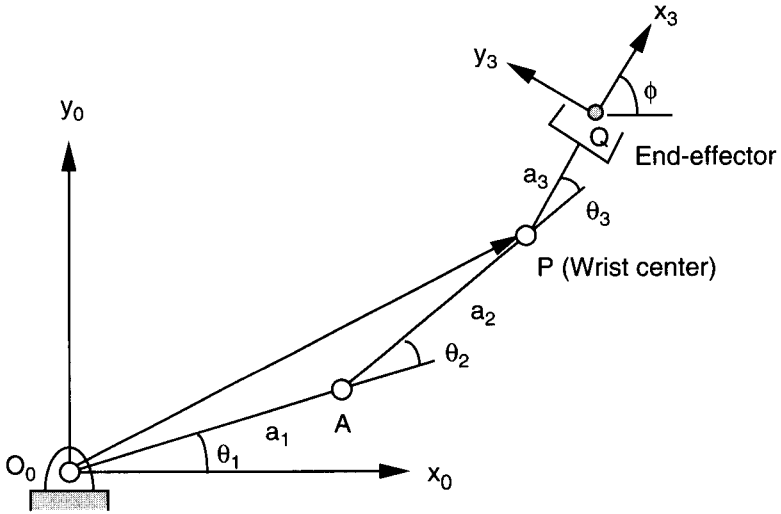


FIGURE 2.7. Vector loop of the planar 3-dof manipulator.

Taking the x and y components of Eq. (2.45) yields

$$p_x = a_1 c\theta_1 + a_2 c\theta_{12}, \quad (2.46)$$

$$p_y = a_1 s\theta_1 + a_2 s\theta_{12}. \quad (2.47)$$

Note that using the vector-loop method, we have derived Eqs. (2.35) and (2.36) with very little effort.

2.6.2 Position Analysis of the Scorbot Robot

For the Scorbot robot shown in Fig. 2.5, the overall transformation matrix is given by Eq. (2.22). We wish to solve the direct and inverse kinematics problems.

(a) Direct Kinematics. For the direct kinematics problem, we simply substitute the given joint angles into Eq. (2.22) to obtain the end-effector position, (q_x, q_y, q_z) , and the orientation in terms of the three unit vectors (u_x, u_y, u_z) , (v_x, v_y, v_z) , and (w_x, w_y, w_z) .

(b) Inverse Kinematics. For the inverse kinematics problem, only 5 of the 12 parameters associated with the end-effector position vector and rotation matrix can be specified at will. This is because the manipulator has only 5 degrees of freedom. It is obvious that the position vector $\mathbf{q}$ and the approach

vector $\mathbf{w}$ cannot be specified simultaneously, due to the fact that $\mathbf{q}$ and $\mathbf{w}$ together depend only on 4 degrees of freedom of the manipulator. For this exercise we assume that $\mathbf{q}$ and $\mathbf{u}$ are specified and that the other two unit vectors, $\mathbf{v}$ and $\mathbf{w}$, are to be determined after the joint angles are found.

Although Eq. (2.22) can be used to solve the inverse kinematics, in what follows we take a more straightforward approach by multiplying both sides of the loop-closure equation by $({}^0A_1)^{-1}$; that is,

$$({}^0A_1)^{-1} {}^0A_5 = {}^1A_2 {}^2A_3 {}^3A_4 {}^4A_5. \quad (2.48)$$

Equating the first column of Eq. (2.48), we obtain

$$u_x c\theta_1 + u_y s\theta_1 = c\theta_{234} c\theta_5, \quad (2.49)$$

$$-u_z = s\theta_{234} c\theta_5, \quad (2.50)$$

$$-u_x s\theta_1 + u_y c\theta_1 = -s\theta_5. \quad (2.51)$$

Similarly, equating the fourth column of Eq. (2.48), we obtain

$$q_x c\theta_1 + q_y s\theta_1 - a_1 = a_2 c\theta_2 + a_3 c\theta_{23} - d_5 s\theta_{234}, \quad (2.52)$$

$$-q_z + d_1 = a_2 s\theta_2 + a_3 s\theta_{23} + d_5 c\theta_{234}, \quad (2.53)$$

$$-q_x s\theta_1 + q_y c\theta_1 = 0. \quad (2.54)$$

The first joint angle, θ_1 , is obtained immediately from Eq. (2.54):

$$\theta_1 = \tan^{-1} \frac{q_y}{q_x}. \quad (2.55)$$

There are two solutions; that is, if $\theta_1 = \theta_1^*$ is a solution, $\theta_1 = \pi + \theta_1^*$ is also a solution. Once θ_1 is found, two solutions for θ_5 are obtained from Eq. (2.51):

$$\theta_5 = \sin^{-1}(u_x s\theta_1 - u_y c\theta_1). \quad (2.56)$$

That is, if $\theta_5 = \theta_5^*$ is a solution, $\theta_5 = \pi - \theta_5^*$ is also a solution.

Corresponding to each solution set of (θ_1, θ_5) , Eqs. (2.49) and (2.50) produce a unique solution of θ_{234} :

$$\theta_{234} = \text{Atan2} \left[-u_z / c\theta_5, (u_x c\theta_1 + u_y s\theta_1) / c\theta_5 \right]. \quad (2.57)$$

Next, we solve Eqs. (2.52) and (2.53) for θ_2 and θ_3 . Equations (2.52) and (2.53) can be written

$$a_2 c\theta_2 + a_3 c\theta_{23} = k_1, \quad (2.58)$$

$$a_2 s\theta_2 + a_3 s\theta_{23} = k_2. \quad (2.59)$$

where $k_1 = q_x c\theta_1 + q_y s\theta_1 - a_1 + d_5 s\theta_{234}$ and $k_2 = -q_z + d_1 - d_5 c\theta_{234}$.

Summing the squares of Eqs. (2.58) and (2.59) yields

$$a_2^2 + a_3^2 + 2a_2 a_3 c\theta_3 = k_1^2 + k_2^2. \quad (2.60)$$

Hence

$$\theta_3 = \cos^{-1} \frac{k_1^2 + k_2^2 - a_2^2 - a_3^2}{2a_2 a_3}. \quad (2.61)$$

and there are two solutions of θ_3 ; that is, if $\theta_3 = \theta_3^*$ is a solution, $\theta_3 = -\theta_3^*$ is also a solution.

Once θ_3 is known, we can solve θ_2 by expanding Eqs. (2.58) and (2.59) as follows:

$$(a_2 + a_3 c\theta_3) c\theta_2 - (a_3 s\theta_3) s\theta_2 = k_1, \quad (2.62)$$

$$(a_3 s\theta_3) c\theta_2 + (a_2 + a_3 c\theta_3) s\theta_2 = k_2. \quad (2.63)$$

Solving Eqs. (2.62) and (2.63) for $c\theta_2$ and $s\theta_2$ yields

$$c\theta_2 = \frac{k_1(a_2 + a_3 c\theta_3) + k_2 a_3 s\theta_3}{a_2^2 + a_3^2 + 2a_2 a_3 c\theta_3},$$

$$s\theta_2 = \frac{-k_1 a_3 s\theta_3 + k_2(a_2 + a_3 c\theta_3)}{a_2^2 + a_3^2 + 2a_2 a_3 c\theta_3}.$$

Hence, corresponding to each solution set of $(\theta_1, \theta_3, \theta_5, \theta_{234})$, we obtain a unique solution of θ_2 :

$$\theta_2 = \text{Atan2}(s\theta_2, c\theta_2). \quad (2.64)$$

Finally, θ_4 is obtained by

$$\theta_4 = \theta_{234} - \theta_2 - \theta_3. \quad (2.65)$$

We conclude that corresponding to each given end-effector location, there are at most eight inverse kinematics solutions.

2.6.3 Position Analysis of the Fanuc S-900W Robot

Figure 2.8 shows a 6-dof manipulator manufactured by Fanuc. In this manipulator, the first joint axis points up vertically along the z_0 -axis, the second joint axis is perpendicular to the first joint axis with a small offset distance $a_1 = OA$, the third joint axis is parallel to the second with an offset distance $a_2 = AB$, and the fourth joint axis is perpendicular to the third joint axis with a small offset distance $a_3 = BC$. In addition, the last three joint axes intersect one another perpendicularly in sequence at a common point P , which is d_4 distance away from point C . This robot belongs to a special class of manipulators where the last three joint axes intersect at the wrist center. The kinematics problem for this type of manipulators can be partitioned into two subchains: one associated with the first three moving links and the other with the last three moving links. That is, in solving the inverse kinematics problem, the position of the wrist center can be solved independently of the orientation part, therefore reducing the complexity of the problem.

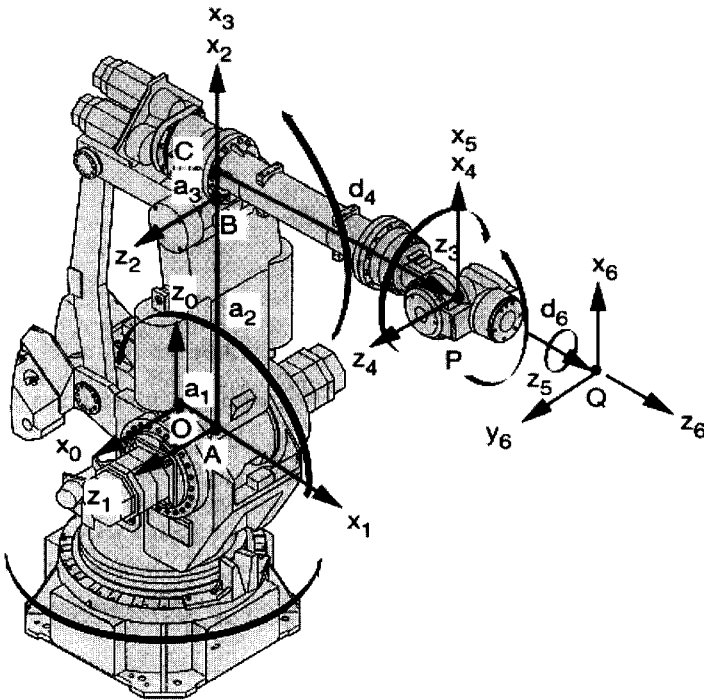


FIGURE 2.8. Fanuc S-900W robot. (Courtesy of Fanuc Robotics North America, Inc., Rochester Hills, Michigan.)

TABLE 2.4. D-H Parameters of the Fanuc S-900W Manipulator

Joint i	α_i	a_i	d_i	θ_i
1	$\pi/2$	a_1	0	θ_1
2	0	a_2	0	θ_2
3	$\pi/2$	a_3	0	θ_3
4	$-\pi/2$	0	d_4	θ_4
5	$\pi/2$	0	0	θ_5
6	0	0	d_6	θ_6

We note that this manipulator employs a four-bar linkage to drive the third joint. The four-bar linkage simply transmits the motion of the third motor mounted on the waist to the third joint. Otherwise, it has no effect on the kinematics of the manipulator. In the following analysis we neglect the effect of the four-bar linkage and treat the manipulator as a serial manipulator.

Using the coordinate systems established in Fig. 2.8, the corresponding link parameters are listed in Table 2.4. Substituting the D-H link parameters into Eq. (2.2), we obtain the D-H transformation matrices:

$${}^0A_1 = \begin{bmatrix} c\theta_1 & 0 & s\theta_1 & a_1c\theta_1 \\ s\theta_1 & 0 & -c\theta_1 & a_1s\theta_1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.66)$$

$${}^1A_2 = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_2c\theta_2 \\ s\theta_2 & c\theta_2 & 0 & a_2s\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.67)$$

$${}^2A_3 = \begin{bmatrix} c\theta_3 & 0 & s\theta_3 & a_3c\theta_3 \\ s\theta_3 & 0 & -c\theta_3 & a_3s\theta_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.68)$$

$${}^3A_4 = \begin{bmatrix} c\theta_4 & 0 & -s\theta_4 & 0 \\ s\theta_4 & 0 & c\theta_4 & 0 \\ 0 & -1 & 0 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.69)$$

$${}^4A_5 = \begin{bmatrix} c\theta_5 & 0 & s\theta_5 & 0 \\ s\theta_5 & 0 & -c\theta_5 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.70)$$

$${}^5A_6 = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ s\theta_6 & c\theta_6 & 0 & 0 \\ 0 & 0 & 1 & d_6 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.71)$$

The end-effector location is given by

$${}^0A_6 = \begin{bmatrix} u_x & v_x & w_x & q_x \\ u_y & v_y & w_y & q_y \\ u_z & v_z & w_z & q_z \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.72)$$

The loop-closure equation is obtained in two steps. First, we multiply Eqs. (2.66), (2.67), and (2.68):

$$\begin{aligned} {}^0A_3 &= {}^0A_1 {}^1A_2 {}^2A_3 \\ &= \begin{bmatrix} c\theta_1 c\theta_{23} & s\theta_1 & c\theta_1 s\theta_{23} & c\theta_1 (a_1 + a_2 c\theta_2 + a_3 c\theta_{23}) \\ s\theta_1 c\theta_{23} & -c\theta_1 & s\theta_1 s\theta_{23} & s\theta_1 (a_1 + a_2 c\theta_2 + a_3 c\theta_{23}) \\ s\theta_{23} & 0 & -c\theta_{23} & a_2 s\theta_2 + a_3 s\theta_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (2.73)$$

Next, we multiply Eqs. (2.69), (2.70), and (2.71):

$$\begin{aligned} {}^3A_6 &= {}^3A_4 {}^4A_5 {}^5A_6 \\ &= \begin{bmatrix} c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6 & -c\theta_4 c\theta_5 s\theta_6 - s\theta_4 c\theta_6 & c\theta_4 s\theta_5 & d_6 c\theta_4 s\theta_5 \\ s\theta_4 c\theta_5 c\theta_6 + c\theta_4 s\theta_6 & -s\theta_4 c\theta_5 s\theta_6 + c\theta_4 c\theta_6 & s\theta_4 s\theta_5 & d_6 s\theta_4 s\theta_5 \\ -s\theta_5 c\theta_6 & s\theta_5 s\theta_6 & c\theta_5 & d_4 + d_6 c\theta_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (2.74)$$

Hence the resulting transformation matrix is given by

$${}^0A_6 = {}^0A_3 {}^3A_6, \quad (2.75)$$

where 0A_6 describes the end effector location.

Substituting Eqs. (2.73) and (2.74) into (2.75) yields the elements of 0A_6 as follows:

$$u_x = c\theta_1 [c\theta_{23}(c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6) - s\theta_{23}s\theta_5 c\theta_6] + s\theta_1 (s\theta_4 c\theta_5 c\theta_6 + c\theta_4 s\theta_6),$$

$$u_y = s\theta_1 [c\theta_{23}(c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6) - s\theta_{23}s\theta_5 c\theta_6] - c\theta_1 (s\theta_4 c\theta_5 c\theta_6 + c\theta_4 s\theta_6),$$

$$u_z = s\theta_{23}(c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6) + c\theta_{23}s\theta_5 c\theta_6,$$

$$\begin{aligned}
v_x &= c\theta_1[-c\theta_{23}(c\theta_4c\theta_5s\theta_6 + s\theta_4c\theta_6) + s\theta_{23}s\theta_5s\theta_6] \\
&\quad + s\theta_1(-s\theta_4c\theta_5s\theta_6 + c\theta_4c\theta_6), \\
v_y &= s\theta_1[-c\theta_{23}(c\theta_4c\theta_5s\theta_6 + s\theta_4c\theta_6) + s\theta_{23}s\theta_5s\theta_6] \\
&\quad - c\theta_1(-s\theta_4c\theta_5s\theta_6 + c\theta_4c\theta_6), \\
v_z &= -s\theta_{23}(c\theta_4c\theta_5s\theta_6 + s\theta_4c\theta_6) - c\theta_{23}s\theta_5s\theta_6, \\
w_x &= c\theta_1(c\theta_{23}c\theta_4s\theta_5 + s\theta_{23}c\theta_5) + s\theta_1s\theta_4s\theta_5, \\
w_y &= s\theta_1(c\theta_{23}c\theta_4s\theta_5 + s\theta_{23}c\theta_5) - c\theta_1s\theta_4s\theta_5, \\
w_z &= s\theta_{23}c\theta_4s\theta_5 - c\theta_{23}c\theta_5, \\
q_x &= c\theta_1[a_1 + a_2c\theta_2 + a_3c\theta_{23} + d_4s\theta_{23} + d_6(c\theta_{23}c\theta_4s\theta_5 + s\theta_{23}c\theta_5)] \\
&\quad + d_6s\theta_1s\theta_4s\theta_5, \\
q_y &= s\theta_1[a_1 + a_2c\theta_2 + a_3c\theta_{23} + d_4s\theta_{23} + d_6(c\theta_{23}c\theta_4s\theta_5 + s\theta_{23}c\theta_5)] \\
&\quad - d_6c\theta_1s\theta_4s\theta_5, \\
q_z &= a_2s\theta_2 + a_3s\theta_{23} - d_4c\theta_{23} + d_6(s\theta_{23}c\theta_4s\theta_5 - c\theta_{23}c\theta_5).
\end{aligned}$$

Although the equations above can be used to solve the inverse kinematics, they are highly nonlinear and difficult to solve. In what follows we present a more efficient method of solution by separating the wrist-center-position problem from the orientation problem.

(a) Wrist Center Position. Note that the last three joint axes intersect at the wrist center point P as shown in Fig. 2.8. Hence rotations of the last three joints do not affect the position of P . Figure 2.9 shows the end-effector coordinate system (x_6, y_6, z_6) , the wrist center P , and the vector relation between them.

The wrist center position with respect to and expressed in the end-effector coordinate system is

$${}^6\mathbf{p} = \overline{QP} = [0, 0, -d_6, 1]^T. \quad (2.76)$$

The wrist center position with respect to and expressed in the base coordinate system is

$${}^0\mathbf{p} = \overline{OP} = \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = \begin{bmatrix} q_x - d_6w_x \\ q_y - d_6w_y \\ q_z - d_6w_z \\ 1 \end{bmatrix}. \quad (2.77)$$

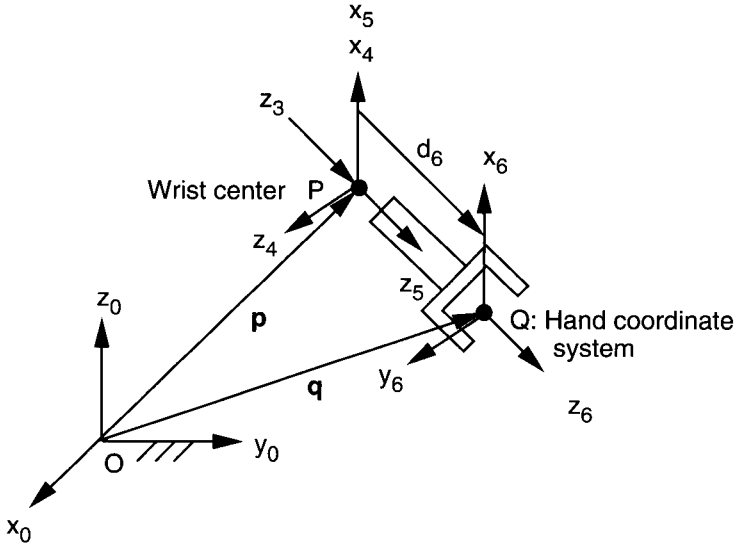


FIGURE 2.9. Hand coordinate system and wrist center position.

Hence, given the end-effector location, we can find the position of the wrist center point P with respect to the base coordinate system. Furthermore, we observe from Fig. 2.8 that the position of the wrist center P with respect to the link 3 coordinate system is given by

$${}^3\mathbf{p} = \overline{CP} = [0, 0, d_4, 1]^T. \quad (2.78)$$

Transforming ${}^3\mathbf{p}$ into the base coordinate system, we obtain

$${}^0\mathbf{p} = {}^0A_3 {}^3\mathbf{p}. \quad (2.79)$$

Equation (2.79) consists of three scalar equations in three unknowns. Hence the position and orientation of the inverse kinematics problem are decoupled.

Theoretically, we can solve Eq. (2.79) for the three joint angles. In what follows we take a simpler approach. Multiplying both sides Eq. (2.79) by the inverse of 0A_1 , we obtain

$$({}^0A_1)^{-1} {}^0\mathbf{p} = {}^1A_3 {}^3\mathbf{p}. \quad (2.80)$$

Substituting Eqs. (2.66) through (2.68) into (2.80) yields

$$p_x c\theta_1 + p_y s\theta_1 - a_1 = a_2 c\theta_2 + a_3 c\theta_{23} + d_4 s\theta_{23}, \quad (2.81)$$

$$p_z = a_2 s\theta_2 + a_3 s\theta_{23} - d_4 c\theta_{23}, \quad (2.82)$$

$$p_x s\theta_1 - p_y c\theta_1 = 0, \quad (2.83)$$

where p_x , p_y , and p_z are given by Eq. (2.77).

A solution for θ_1 is found immediately by solving Eq. (2.83).

$$\theta_1 = \tan^{-1} \frac{p_y}{p_x}. \quad (2.84)$$

Hence there are two solutions of θ_1 . Specifically, if $\theta_1 = \theta_1^*$ is a solution, $\theta_1 = \theta_1^* + \pi$ is also a solution, where $\pi \geq \theta_1^* \geq 0$. We call $\theta_1 = \theta_1^*$ the front-reach solution and $\theta_1 = \theta_1^* + \pi$ the back-reach solution. Because of the four-bar linkage and other mechanical constraints, the back-reach solution is physically impossible.

An observation of the kinematic structure reveals that the distance between point A and the wrist center P is independent of θ_1 and θ_2 , which implies that these two variables can be eliminated simultaneously. Summing the squares of Eqs. (2.81), (2.82), and (2.83), gives

$$\kappa_1 s\theta_3 + \kappa_2 c\theta_3 = \kappa_3, \quad (2.85)$$

where $\kappa_1 = 2a_2d_4$, $\kappa_2 = 2a_2a_3$, and $\kappa_3 = p_x^2 + p_y^2 + p_z^2 - 2p_xa_1c\theta_1 - 2p_ya_1s\theta_1 + a_1^2 - a_2^2 - a_3^2 - d_4^2$.

We can convert Eq. (2.85) into a polynomial by making use of the following trigonometric identities:

$$c\theta_3 = \frac{1 - t_3^2}{1 + t_3^2} \quad \text{and} \quad s\theta_3 = \frac{2t_3}{1 + t_3^2}, \quad \text{where} \quad t_3 = \tan \frac{\theta_3}{2}.$$

Substituting the trigonometric identities above into Eq. (2.85) yields

$$(\kappa_3 + \kappa_2)t_3^2 - 2\kappa_1t_3 + (\kappa_3 - \kappa_2) = 0. \quad (2.86)$$

Hence

$$\frac{\theta_3}{2} = \tan^{-1} \frac{\kappa_1 \pm \sqrt{\kappa_1^2 + \kappa_2^2 - \kappa_3^2}}{\kappa_3 + \kappa_2}. \quad (2.87)$$

Equation (2.86) yields (1) two real roots if $\kappa_1^2 + \kappa_2^2 - \kappa_3^2 > 0$, (2) one double root if $\kappa_1^2 + \kappa_2^2 - \kappa_3^2 = 0$, and (3) no real roots if $\kappa_1^2 + \kappa_2^2 - \kappa_3^2 < 0$. When Eq. (2.86) yields a double root, the arm is either in a fully stretched or a folded-back configuration. On the other hand, if Eq. (2.86) yields no real roots, the position is not reachable. Figure 2.10 shows two different arm configurations, corresponding to the two solutions of θ_3 .

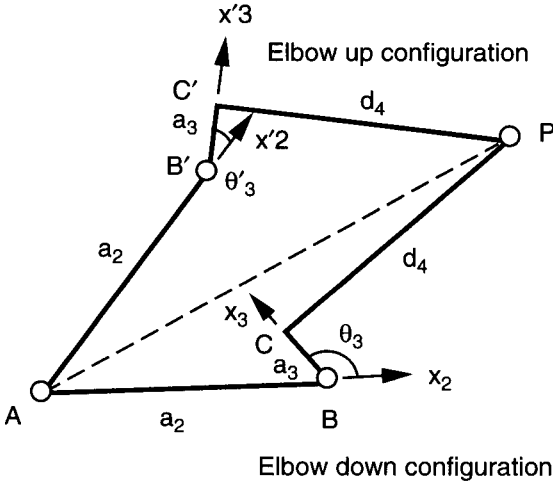


FIGURE 2.10. Two different arm configurations.

Once θ_1 and θ_3 are known, θ_2 can be obtained by back substitution. Expanding Eqs. (2.81) and (2.82), we obtain

$$\mu_1 c\theta_2 + \nu_1 s\theta_2 = \gamma_1, \quad (2.88)$$

$$\mu_2 c\theta_2 + \nu_2 s\theta_2 = \gamma_2, \quad (2.89)$$

where

$$\mu_1 = a_2 + a_3 c\theta_3 + d_4 s\theta_3,$$

$$\nu_1 = -a_3 s\theta_3 + d_4 c\theta_3,$$

$$\gamma_1 = p_x c\theta_1 + p_y s\theta_1 - a_1,$$

$$\mu_2 = a_3 s\theta_3 - d_4 c\theta_3,$$

$$\nu_2 = a_2 + a_3 c\theta_3 + d_4 s\theta_3,$$

$$\gamma_2 = p_z.$$

Therefore, we can solve Eqs. (2.88) and (2.89) for $c\theta_2$ and $s\theta_2$. Once $s\theta_2$ and $c\theta_2$ are found, a unique value of θ_2 is obtained by taking

$$\theta_2 = \text{Atan2}(s\theta_2, c\theta_2). \quad (2.90)$$

We conclude that given the wrist center position, mathematically there are at most four possible arm configurations, but due to the mechanical limits, only two are physically possible.

(b) End-Effector Orientation. Once θ_1 , θ_2 , and θ_3 are solved, 0A_3 is completely known. The remaining joint angles can be found by multiplying both sides of Eq. (2.75) by $({}^0A_3)^{-1}$:

$${}^3A_6 = ({}^0A_3)^{-1} {}^0A_6. \quad (2.91)$$

We note that the elements on the right-hand side of Eq. (2.91) are known, and only the rotation part of Eq. (2.91) is needed for computation of the last three joint angles. The rotation matrices 0R_3 and 3R_6 are given by the upper 3×3 submatrices of Eqs. (2.73) and (2.74), respectively.

Equating the 3×3 element of Eq. (2.91) yields

$$\theta_5 = \cos^{-1} r_{33}, \quad (2.92)$$

where $r_{33} = w_x c\theta_1 s\theta_{23} + w_y s\theta_1 s\theta_{23} - w_z c\theta_{23}$. Hence, corresponding to each solution set of θ_1 , θ_2 , and θ_3 , Eq. (2.92) yields (1) two real roots if $|r_{33}| < 1$, and (2) $\theta_5 = 0$ or π if $|r_{33}| = 1$. When $\theta_5 = 0$ or π , the sixth joint axis, z_5 , is in line with the fourth joint axis, z_3 , and the wrist is said to be in a *singular configuration*. The condition $|r_{33}| > 1$ cannot physically arise.

Assuming that $s\theta_5 \neq 0$, we can solve θ_4 and θ_6 as follows. Equating the 1×3 element of Eq. (2.91) yields

$$c\theta_4 = \frac{w_x c\theta_1 c\theta_{23} + w_y s\theta_1 c\theta_{23} + w_z s\theta_{23}}{s\theta_5}. \quad (2.93)$$

Equating the 2×3 element of Eq. (2.91) yields

$$s\theta_4 = \frac{w_x s\theta_1 - w_y c\theta_1}{s\theta_5}. \quad (2.94)$$

Hence, corresponding to each solution set of θ_1 , θ_2 , θ_3 , and θ_5 , Eqs. (2.93) and (2.94) yield a unique solution of θ_4 :

$$\theta_4 = \text{Atan2}(s\theta_4, c\theta_4). \quad (2.95)$$

Similarly, equating the 3×1 element of Eq. (2.91) yields

$$c\theta_6 = -\frac{u_x c\theta_1 s\theta_{23} + u_y s\theta_1 s\theta_{23} - u_z c\theta_{23}}{s\theta_5}. \quad (2.96)$$

Equating the 3×2 element of Eq. (2.91) yields

$$s\theta_6 = \frac{v_x c\theta_1 s\theta_{23} + v_y s\theta_1 s\theta_{23} - v_z c\theta_{23}}{s\theta_5}. \quad (2.97)$$

Hence, corresponding to each solution set of $\theta_1, \theta_2, \theta_3, \theta_4$, and θ_5 , Eqs. (2.96) and (2.97) yield a unique solution of θ_6 :

$$\theta_6 = \text{Atan2}(s\theta_6, c\theta_6). \quad (2.98)$$

We conclude that corresponding to each solution set of the first three joint angles, there are two possible wrist configurations. Since there are four possible upper arm configurations, a total of eight manipulator postures are possible. However, due to mechanical limits, fewer than eight manipulator postures are physically realizable. When $s\theta_5 = 0$, Eqs. (2.93) through (2.98) degenerate. For such a singular condition, only the sum or difference of θ_4 and θ_6 can be computed.

2.6.4 Tsai and Morgan's Solution

In this section we outline a solution method developed by Tsai and Morgan (1985), who reduced the problem to a system of four equations and then employed a numerical method, known as the *homotopy method*, to find all solutions to the inverse kinematics of a general 6R manipulator. They also derived closed-form solutions for manipulators in which three consecutive joint axes either intersect at a common point or are parallel to one another.

Figure 2.11 shows a general 6R manipulator where point Q denotes the origin and $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ denote three orthogonal unit vectors of the end effector coordinate system. Using the Denavit-Hartenberg method, a loop-closure equation can be written as

$${}^0A_1 {}^1A_2 {}^2A_3 {}^3A_4 {}^4A_5 {}^5A_6 = {}^0A_6. \quad (2.99)$$

For convenience, we introduce a position vector $\mathbf{p}$ of the origin and a unit vector $\mathbf{e}$ of the z_5 -axis of the fifth coordinate system as shown in Fig. 2.11. These two vectors can be expressed in the fifth coordinate system as ${}^5\mathbf{p} = [0, 0, 0, 1]^T$ and ${}^5\mathbf{e} = [0, 0, 1, 0]^T$, or in the fixed coordinate system as $\mathbf{p} = {}^0\mathbf{p} = [p_x, p_y, p_z, 1]^T$ and $\mathbf{e} = {}^0\mathbf{e} = [e_x, e_y, e_z, 0]^T$. Since both the point P and the vector $\mathbf{e}$ are attached to the end effector, $\mathbf{p}$ and $\mathbf{e}$ can be computed from the given end-effector location as follows:

$$\begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = {}^0A_6 {}^6A_5 \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -u_x a_6 - (v_x s\alpha_6 + w_x c\alpha_6)d_6 + q_x \\ -u_y a_6 - (v_y s\alpha_6 + w_y c\alpha_6)d_6 + q_y \\ -u_z a_6 - (v_z s\alpha_6 + w_z c\alpha_6)d_6 + q_z \\ 1 \end{bmatrix}, \quad (2.100)$$

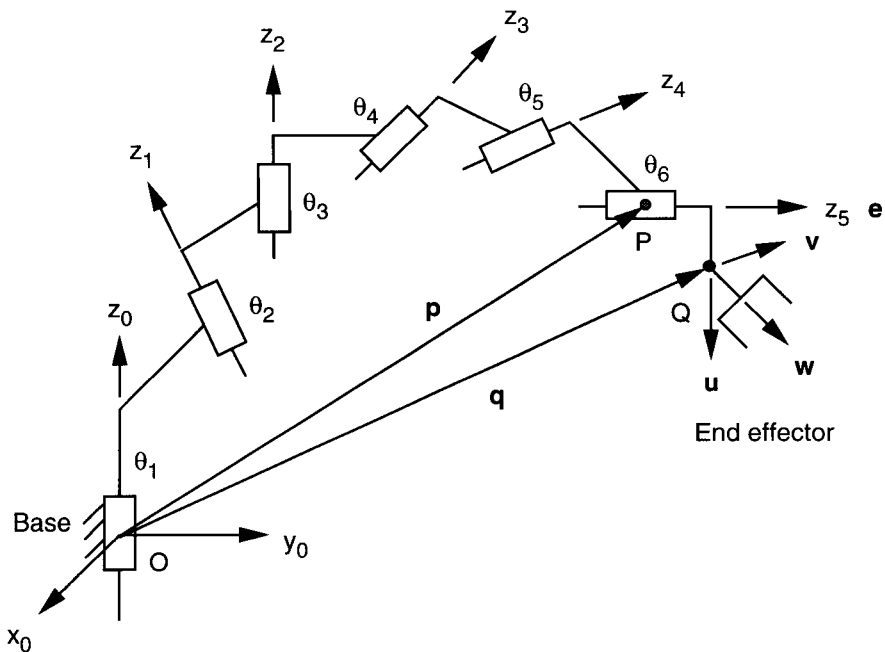


FIGURE 2.11. General 6R manipulator.

$$\begin{bmatrix} e_x \\ e_y \\ e_z \\ 0 \end{bmatrix} = {}^0A_6 {}^6A_5 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} v_x \alpha_6 + w_x c\alpha_6 \\ v_y \alpha_6 + w_y c\alpha_6 \\ v_z \alpha_6 + w_z c\alpha_6 \\ 0 \end{bmatrix}. \quad (2.101)$$

Equations (2.100) and (2.101) imply that once the end-effector location is given, the point P and the direction of z_5 -axis can be found.

The transformation between ${}^5\mathbf{p}$ and ${}^0\mathbf{p}$ and between ${}^5\mathbf{e}$ and ${}^0\mathbf{e}$ can be written as

$${}^0\mathbf{p} = {}^0A_1 {}^1A_2 {}^2A_3 {}^3A_4 {}^4A_5 {}^5\mathbf{p}, \quad (2.102)$$

$${}^0\mathbf{e} = {}^0A_1 {}^1A_2 {}^2A_3 {}^3A_4 {}^4A_5 {}^5\mathbf{e}. \quad (2.103)$$

To simplify the analysis, we multiply both sides of Eqs. (2.102) and (2.103) by $({}^0A_1 {}^1A_2)^{-1}$. The resulting equations can be written

$${}^2\mathbf{p} = {}^2\mathbf{p}', \quad (2.104)$$

$${}^2\mathbf{e} = {}^2\mathbf{e}', \quad (2.105)$$

where

$$\begin{aligned} {}^2\mathbf{p} &= {}^2A_3 {}^3A_4 {}^4A_5 {}^5\mathbf{p}, \\ {}^2\mathbf{p}' &= ({}^1A_2)^{-1} ({}^0A_1)^{-1} {}^0\mathbf{p}, \\ {}^2\mathbf{e} &= {}^2A_3 {}^3A_4 {}^4A_5 {}^5\mathbf{e}, \\ {}^2\mathbf{e}' &= ({}^1A_2)^{-1} ({}^0A_1)^{-1} {}^0\mathbf{e} \end{aligned}$$

are the position vectors of P and the direction of the z_5 -axis with reference to the (x_2, y_2, z_2) coordinate system.

Equations (2.104) and (2.105) constitute a set of six scalar equations free of the variable θ_6 . However, only two of the three scalar equations in Eq. (2.105) are independent, because the components of $\mathbf{e}$ must satisfy the condition of a unit vector. Hence there are only five independent equations in five unknowns, $\theta_1, \dots, \theta_5$. The x and y components in Eqs. (2.104) and (2.105) are third-degree polynomials, while the z -component is a second-degree polynomial in the sines and cosines of five joint angles. We note that by using this approach, θ_6 does not appear in the system of equations and therefore reduces the complexity of the problem. In the following we eliminate θ_3 from the system of equations above.

Elimination of θ_3 . First, we notice that both z -components of Eqs. (2.104) and (2.105) are already free of the variable θ_3 . Expanding the z -components of Eqs. (2.104) and (2.105), yields

$$\begin{aligned} h_x \alpha_2 s \theta_2 - h_y \alpha_2 c \theta_2 - g_x \alpha_3 s \theta_4 + g_y \alpha_3 c \theta_4 \\ = -h_z \alpha_2 + d_2 \alpha_2 + g_z \alpha_3 + d_3, \end{aligned} \quad (2.106)$$

$$\begin{aligned} n_x \alpha_2 s \theta_2 - n_y \alpha_2 c \theta_2 - m_x \alpha_3 s \theta_4 + m_y \alpha_3 c \theta_4 \\ = -n_z \alpha_2 + m_z \alpha_3, \end{aligned} \quad (2.107)$$

where

$$\begin{aligned} g_x &= a_5 c \theta_5 + a_4, \\ g_y &= -a_5 \alpha_4 s \theta_5 + d_5 \alpha_4, \\ g_z &= a_5 \alpha_4 s \theta_5 + d_5 \alpha_4 + d_4, \\ h_x &= p_x c \theta_1 + p_y s \theta_1 - a_1, \\ h_y &= -p_x \alpha_1 s \theta_1 + p_y \alpha_1 c \theta_1 + (p_z - d_1) \alpha_1, \\ h_z &= p_x \alpha_1 s \theta_1 - p_y \alpha_1 c \theta_1 + (p_z - d_1) \alpha_1, \\ m_x &= \alpha_5 s \theta_5, \end{aligned}$$

$$\begin{aligned}
m_y &= c\alpha_4 s\alpha_5 c\theta_5 + s\alpha_4 c\alpha_5, \\
m_z &= -s\alpha_4 s\alpha_5 c\theta_5 + c\alpha_4 c\alpha_5, \\
n_x &= e_x c\theta_1 + e_y s\theta_1, \\
n_y &= -e_x c\alpha_1 s\theta_1 + e_y c\alpha_1 c\theta_1 + e_z s\alpha_1, \\
n_z &= e_x s\alpha_1 s\theta_1 - e_y s\alpha_1 c\theta_1 + e_z c\alpha_1.
\end{aligned}$$

A third equation that is free of θ_3 is obtained by performing the dot product ${}^2\mathbf{p} \cdot {}^2\mathbf{e} = {}^2\mathbf{p}' \cdot {}^2\mathbf{e}'$. Substituting Eqs. (2.104) and (2.105) into the dot product and simplifying, we obtain

$$\begin{aligned}
&a_2 n_y s\theta_2 + a_2 n_x c\theta_2 + (a_3 m_y + d_3 m_x s\alpha_3) s\theta_4 + (a_3 m_x - d_3 m_y s\alpha_3) c\theta_4 \\
&= -a_1 n_x - d_2 n_z - a_4 m_x - m_z (d_3 c\alpha_3 + d_4) + k_1,
\end{aligned} \tag{2.108}$$

where $k_1 = -d_5 c\alpha_5 + p_x e_x + p_y e_y + (p_z - d_1) e_z$.

A fourth equation that is free of θ_3 is obtained by equating the sum of the squares of the x , y , and z components on both sides of Eq. (2.104). Expanding $({}^2\mathbf{p})^2 = ({}^2\mathbf{p}')^2$ yields

$$\begin{aligned}
&a_2 h_y s\theta_2 + a_2 h_x c\theta_2 + (a_3 g_y + d_3 g_x s\alpha_3) s\theta_4 + (a_3 g_x - d_3 g_y s\alpha_3) c\theta_4 \\
&= -a_1 h_x - d_2 h_z - a_4 g_x - g_z (d_3 c\alpha_3 + d_4) + k_2,
\end{aligned} \tag{2.109}$$

where

$$k_2 = 0.5[p_x^2 + p_y^2 + (p_z - d_1)^2 - a_1^2 + a_2^2 + d_2^2 - a_3^2 - d_3^2 + a_4^2 + d_4^2 - a_5^2 - d_5^2].$$

Equations (2.106) through (2.109) represent a system of four second-degree polynomials in the sines and cosines of four joint angles. We may consider $\sin \theta_i$ and $\cos \theta_i$ as two independent variables and add the following trigonometric identities as supplementary equations of constraint:

$$s^2\theta_i + c^2\theta_i = 1, \quad \text{for } i = 1, 2, 4, 5. \tag{2.110}$$

In this way, we obtain a system of eight second-degree polynomials in eight variables. Tsai and Morgan employed a continuation method to solve the system of equations above and showed that the most general 6-dof, 6R robot has at most 16 significant solutions. See Appendix A for more details.

The system of equations will decouple when any three consecutive joint axes either intersect at a common point or are parallel to one another. For these special geometries, closed-form solutions can be derived. In what follows we illustrate the decoupling by solving the inverse kinematics of two special cases that are most commonly implemented in industrial robots. Other special cases can be derived by applying the kinematic inversions.

(a) Last Three Joint Axes Intersecting at a Common Point. When the last three joint axes intersect at a common point, $a_4 = a_5 = d_5 = 0$ identically. Substituting these values into Eqs. (2.106) and (2.109) yields

$$h_x s\theta_2 - h_y c\theta_2 = \mu_1 / s\alpha_2, \quad (2.111)$$

$$h_y s\theta_2 + h_x c\theta_2 = \mu_2 / a_2, \quad (2.112)$$

provided that $s\alpha_2 \neq 0$ and $a_2 \neq 0$, where

$$\mu_1 = -h_z c\alpha_2 + d_2 c\alpha_2 + d_3 + d_4 c\alpha_3,$$

$$\begin{aligned} \mu_2 = & -a_1 h_x - d_2 h_z - d_3 d_4 c\alpha_3 \\ & + 0.5[p_x^2 + p_y^2 + (p_z - d_1)^2 - a_1^2 + a_2^2 + d_2^2 - a_3^2 - d_3^2 - d_4^2]. \end{aligned}$$

Equations (2.111) and (2.112) contain only two unknown variables and hence are completely decoupled from Eqs. (2.107) and (2.108). To eliminate θ_2 , we sum the squares of Eqs. (2.111) and (2.112).

$$h_x^2 + h_y^2 = (\mu_1 / s\alpha_2)^2 + (\mu_2 / a_2)^2. \quad (2.113)$$

Equation (2.113) contains only one variable, θ_1 . We may convert it into a fourth-degree polynomial in t_1 by replacing $s\theta_1$ with $2t_1/(1 + t_1^2)$ and $c\theta_1$ with $(1 - t_1^2)/(1 + t_1^2)$, where $t_1 = \tan(\theta_1/2)$. Hence, for each given end-effector position, there are at most four real solutions of θ_1 . Once θ_1 is found, a unique solution of θ_2 can be obtained by solving Eqs. (2.111) and (2.112) simultaneously for $s\theta_2$ and $c\theta_2$, and then applying the two-argument arctangent function. Following that, a unique solution of θ_3 can be found by solving the two scalar equations associated with the x and y components of Eq. (2.104).

Corresponding to each solution set of $(\theta_1, \theta_2, \theta_3)$, two sets of $(\theta_4, \theta_5, \theta_6)$ can be found by following the procedure outlined in the earlier example. Hence we conclude that there are at most eight possible solutions sets (manipulator postures).

(b) Joint Axes 2, 3, and 4 Parallel to One Another. When the second, third, and fourth joint axes are parallel to one another, $\alpha_2 = \alpha_3 = 0$ identically. Further, since the common normals between the second and third joint axes and between the third and fourth joint axes are indeterminate, we can always define these two common normals such that $d_2 = d_3 = 0$. With these values, Eqs. (2.106) and (2.107) reduce to

$$s\alpha_4 s\theta_5 = \frac{h_z - d_5 c\alpha_4 - d_4}{a_5}, \quad (2.114)$$

$$s\alpha_4 c\theta_5 = \frac{-n_z + c\alpha_4 c\alpha_5}{s\alpha_5}, \quad (2.115)$$

provided that $a_5 \neq 0$ and $s\alpha_5 \neq 0$.

Again, Eqs. (2.114) and (2.115) contain only two unknown variables and hence are completely decoupled from Eqs. (2.108) and (2.109). We can eliminate θ_5 by summing the squares of Eq. (2.114) and (2.115):

$$s^2\alpha_4 = \left(\frac{h_z - d_5 c\alpha_4 - d_4}{a_5} \right)^2 + \left(\frac{-n_z + c\alpha_4 c\alpha_5}{s\alpha_5} \right)^2. \quad (2.116)$$

Equation (2.116) contains only one variable, θ_1 . We may convert it into a fourth-degree polynomial in t_1 by replacing $s\theta_1$ with $2t_1/(1+t_1^2)$ and $c\theta_1$ with $(1-t_1^2)/(1+t_1^2)$, where $t_1 = \tan(\theta_1/2)$. Hence for each given end-effector position and orientation, there are at most four real solutions of θ_1 . Once θ_1 is known, a unique solution of θ_5 can be obtained by solving Eqs. (2.114) and (2.115) simultaneously for $s\theta_5$ and $c\theta_5$ and then applying the two-argument arctangent function.

When $\alpha_2 = \alpha_3 = 0$, the two scalar equations corresponding to the x and y components of Eq. (2.105) reduce to

$$m_x c\theta_{34} + m_y s\theta_{34} = n_x c\theta_2 + n_y s\theta_2, \quad (2.117)$$

$$m_x s\theta_{34} - m_y c\theta_{34} = -n_x s\theta_2 + n_y c\theta_2. \quad (2.118)$$

Equations (2.117) and (2.118) contain two unknown variables, θ_{34} and θ_2 . We may reduce these two equations to a single equation in one variable by the following procedure. Subtracting Eq. (2.118) $\times s\theta_2$ from (2.117) $\times c\theta_2$ yields

$$m_x c\theta_{234} + m_y s\theta_{234} = n_x. \quad (2.119)$$

Adding Eq. (2.118) $\times c\theta_2$ to (2.117) $\times s\theta_2$ yields

$$m_x s\theta_{234} - m_y c\theta_{234} = n_y. \quad (2.120)$$

Hence, corresponding to each solution set of θ_1 and θ_5 , a unique solution of θ_{234} can be obtained by solving Eqs. (2.119) and (2.120) for $s\theta_{234}$ and $c\theta_{234}$, and then applying the two-argument arctangent function.

Similarly, the two scalar equations corresponding to the x and y components of Eq. (2.104) reduce to

$$g_x c\theta_{34} + g_y s\theta_{34} + a_3 c\theta_3 = h_x c\theta_2 + h_y s\theta_2 - a_2, \quad (2.121)$$

$$g_x s\theta_{34} - g_y c\theta_{34} + a_3 s\theta_3 = -h_x s\theta_2 + h_y c\theta_2. \quad (2.122)$$

Subtracting Eq. (2.122) $\times s\theta_2$ from (2.121) $\times c\theta_2$ yields

$$g_x c\theta_{234} + g_y s\theta_{234} + a_3 c\theta_{23} = h_x - a_2 c\theta_2. \quad (2.123)$$

Adding Eq. (2.122) $\times c\theta_2$ to (2.121) $\times s\theta_2$ yields

$$g_x s\theta_{234} - g_y c\theta_{234} + a_3 s\theta_{23} = h_y - a_2 s\theta_2. \quad (2.124)$$

Summing the squares of [Eq. (2.123) - h_x] and [Eq. (2.124) - h_y] yields

$$2a_3(k_3 c\theta_{23} + k_4 s\theta_{23}) + k_5 = 0, \quad (2.125)$$

where

$$k_3 = g_x c\theta_{234} + g_y s\theta_{234} - h_x,$$

$$k_4 = g_x s\theta_{234} - g_y c\theta_{234} - h_y,$$

$$k_5 = -a_2^2 + a_3^2 + k_3^2 + k_4^2.$$

Hence, corresponding to each solution set of $(\theta_1, \theta_5, \theta_{234})$, Eq. (2.125) yields two solutions of θ_{23} . Once θ_{23} is found, Eqs. (2.123) and (2.124) yield a unique solution of θ_2 . We conclude that there are at most eight possible solution sets.

2.7 METHOD OF SUCCESSIVE SCREW DISPLACEMENTS

In this section we study a method of analysis based on the concept of *successive screw displacements*. First, the transformation matrix associated with a screw displacement is derived. Then the concept of the *resultant screw* of two successive screw displacements is described. Then the concept is applied to the position analysis of serial manipulators.

2.7.1 Transformation Based on Screw Displacement

Chasles' theorem states that the general spatial displacement of a rigid body is a translation plus a rotation. A stronger form of the theorem states that regardless of how a rigid body is displaced from one location to another, the displacement can be regarded as a rotation about and a translation along some axis. Such a combination of translation and rotation is called a *screw displacement* (Bottema and Roth, 1979). In what follows we derive a homogeneous transformation based on the concept of screw displacement.

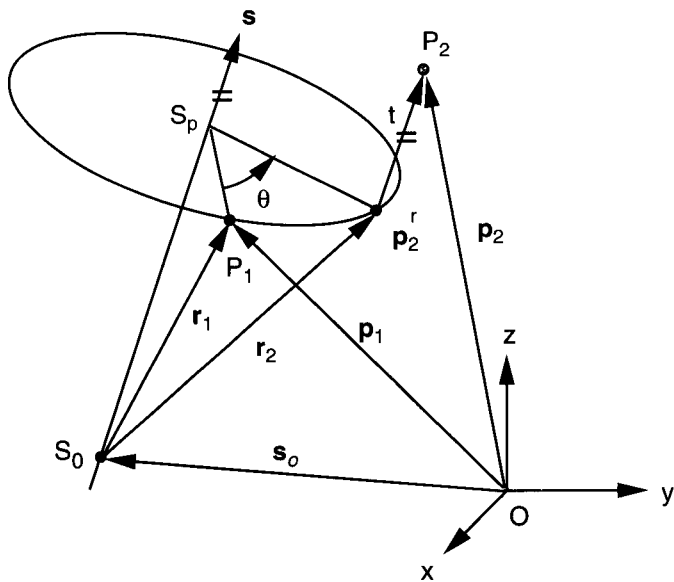


FIGURE 2.12. Vector diagram of a spatial displacement.

Figure 2.12 shows a point P that is displaced from a first position P_1 to a second position P_2 by a rotation of θ about a screw axis followed by a translation of t along the same axis. The rotation brings P from P_1 to P'_1 , and the translation brings P from P'_1 to P_2 . In the figure, $\mathbf{s} = [s_x, s_y, s_z]^T$ denotes a unit vector along the direction of the screw axis, and $\mathbf{s}_o = [s_{ox}, s_{oy}, s_{oz}]^T$ denotes the position vector of a point lying on the screw axis. The rotation angle θ and the translational distance t are called the *screw parameters*. The screw axis together with the screw parameters completely define the general displacement of a rigid body. Note that for a general displacement of a rigid body, the screw axis does not necessarily pass through the origin of the fixed frame.

The displacement equation due to a rotation about an axis passing through the origin was derived in Chapter 1. Hence we only need to take care of the fact that the screw axis does not pass through the origin and add the contribution due to a translation along the screw axis. Referring to Fig. 2.12, we observe that

$$\mathbf{r}_1 = \mathbf{p}_1 - \mathbf{s}_o, \quad (2.126)$$

$$\mathbf{r}_2 = \mathbf{p}_2 - \mathbf{s}_o - t\mathbf{s}. \quad (2.127)$$

Substituting Eqs. (2.126) and (2.127) into (1.35), we obtain

$$\mathbf{p}_2 = \mathbf{s}_o + t\mathbf{s} + (\mathbf{p}_1 - \mathbf{s}_o)c\theta + \mathbf{s} \times (\mathbf{p}_1 - \mathbf{s}_o)s\theta + [(\mathbf{p}_1 - \mathbf{s}_o)^T \mathbf{s}]\mathbf{s}(1 - c\theta). \quad (2.128)$$

Equation (2.128) is known as *Rodrigues's formula* for the general spatial displacement of a rigid body. Expanding Eq. (2.128) and replacing $\mathbf{p}_1$ by ${}^B\mathbf{p}$ and $\mathbf{p}_2$ by ${}^A\mathbf{p}$, we obtain

$${}^A\mathbf{p} = {}^A R_B {}^B\mathbf{p} + {}^A\mathbf{q}, \quad (2.129)$$

where the elements of the rotation matrix, a_{ij} , are given by Eq. (1.37), and the position of the origin, ${}^A\mathbf{q}$, of the moving frame is given by

$$\begin{aligned} q_x &= ts_x - s_{ox}(a_{11} - 1) - s_{oy}a_{12} - s_{oz}a_{13}, \\ q_y &= ts_y - s_{ox}a_{21} - s_{oy}(a_{22} - 1) - s_{oz}a_{23}, \\ q_z &= ts_z - s_{ox}a_{31} - s_{oy}a_{32} - s_{oz}(a_{33} - 1). \end{aligned} \quad (2.130)$$

Equation (2.129) can be written as a homogeneous transformation:

$${}^A\hat{\mathbf{p}} = A {}^B\hat{\mathbf{p}} \quad (2.131)$$

where A is a 4×4 transformation matrix the elements of which are given by

$$\begin{aligned} a_{11} &= (s_x^2 - 1)(1 - c\theta) + 1, \\ a_{12} &= s_x s_y (1 - c\theta) - s_z s\theta, \\ a_{13} &= s_x s_z (1 - c\theta) + s_y s\theta, \\ a_{21} &= s_y s_x (1 - c\theta) + s_z s\theta, \\ a_{22} &= (s_y^2 - 1)(1 - c\theta) + 1, \\ a_{23} &= s_y s_z (1 - c\theta) - s_x s\theta, \\ a_{31} &= s_z s_x (1 - c\theta) - s_y s\theta, \\ a_{32} &= s_z s_y (1 - c\theta) + s_x s\theta, \\ a_{33} &= (s_z^2 - 1)(1 - c\theta) + 1, \\ a_{14} &= ts_x - s_{ox}(a_{11} - 1) - s_{oy}a_{12} - s_{oz}a_{13}, \\ a_{24} &= ts_y - s_{ox}a_{21} - s_{oy}(a_{22} - 1) - s_{oz}a_{23}, \\ a_{34} &= ts_z - s_{ox}a_{31} - s_{oy}a_{32} - s_{oz}(a_{33} - 1), \end{aligned}$$

$$\begin{aligned}
a_{41} &= 0, \\
a_{42} &= 0, \\
a_{43} &= 0, \\
a_{44} &= 1.
\end{aligned} \tag{2.132}$$

The upper left 3×3 submatrix of A represents the rotation of the rigid body. The upper right 3×1 submatrix represents the translation of the origin Q (i.e., $a_{14} = q_x$, $a_{24} = q_y$, and $a_{34} = q_z$). This representation of a spatial displacement requires eight parameters: three associated with the direction of the screw axis, three associated with the location of the screw axis, one associated with the rotation angle, and one associated with the translational distance. However, only two of the three parameters associated with the direction of the screw axis are independent since they must satisfy the condition of a unit vector:

$$\mathbf{s}^T \mathbf{s} = 1. \tag{2.133}$$

Similarly, only two of the three parameters associated with the location of the screw axis are independent, since S_o can be any point on the screw axis. For convenience, we may choose $\mathbf{s}_o$ to be normal to the screw axis:

$$\mathbf{s}_o^T \mathbf{s} = 0. \tag{2.134}$$

Given the screw axis and screw parameters, we can compute the elements of the transformation matrix by Eq. (2.132). On the other hand, given the spatial displacement of a rigid body in terms of a rotation matrix, ${}^A R_B$, and a translation vector, ${}^A \mathbf{q}$, we can compute the screw axis and the screw parameters as follows. The angle of rotation is given by

$$\theta = \cos^{-1} \frac{a_{11} + a_{22} + a_{33} - 1}{2}. \tag{2.135}$$

There are two solutions of θ , one being the negative of the other. Once the rotation angle is known, the direction of the screw axis is computed by

$$\begin{aligned}
s_x &= \frac{a_{32} - a_{23}}{2s\theta}, \\
s_y &= \frac{a_{13} - a_{31}}{2s\theta}, \\
s_z &= \frac{a_{21} - a_{12}}{2s\theta}.
\end{aligned} \tag{2.136}$$

The translational distance is calculated by

$$t = \mathbf{q}^T \mathbf{s}, \quad (2.137)$$

and the screw axis location is obtained by solving any two of the three equations in Eq. (2.130) along with Eq. (2.134). Since these equations are linear, there exists one solution corresponding to each solution set of $\mathbf{s}$, θ , and t .

From the derivation above, it appears that there are two solutions of the screw axis, one being the negative of the other. In reality, these two solutions represent the same screw, since a $-\theta$ rotation about and a $-t$ translation along the $-(\mathbf{s}, s_o)$ screw axis produces the same result as a $+\theta$ rotation about and a $+t$ translation along the $+(\mathbf{s}, s_o)$ screw axis.

2.7.2 Successive Screw Displacements

We now apply the concept of screw displacements to kinematic analysis of open-loop chains. Figure 2.13 shows a rigid body σ which is guided to a fixed base by a dyad that is made up of two kinematic pairs, denoted by $\$1$ and $\$2$, respectively. The first kinematic pair connects the first moving link to the fixed base, and the second kinematic pair connects the second moving link to the fixed base.

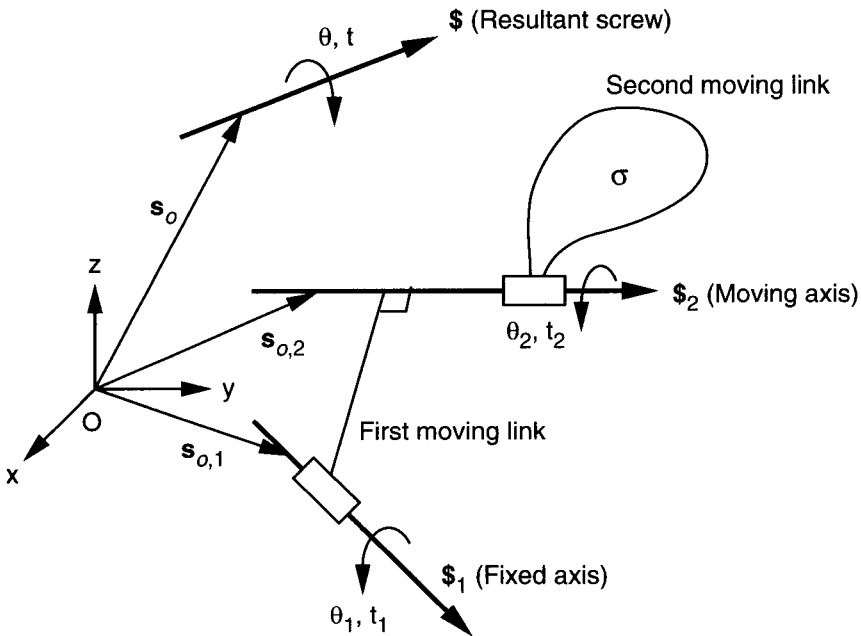


FIGURE 2.13. Two-link chain and its associated screw displacements.

link (σ) to the first. We call the axis of the first kinematic pair the *fixed joint axis* and the axis of the second kinematic pair the *moving joint axis*.

As the rigid body is rotated about and/or translated along these two joint axes, we may think of its displacement as the resultant of a screw displacement of (θ_1, t_1) about the fixed joint axis followed by another displacement of (θ_2, t_2) about the displaced moving joint axis. We may also think of the displacement as the resultant of a screw displacement of (θ_2, t_2) about the moving joint axis in its initial location followed by another displacement of (θ_1, t_1) about the fixed joint axis. As a matter of fact, these two screw displacements can occur in any order or simultaneously as long as the total amounts of rotations and/or translations about these two joint axes are correct. That is, the order in which the joint variables change their values can be arbitrary. The changes can be one at a time in any order or simultaneous.

A difficult scheme is to consider the resultant screw, S , as a screw displacement of S_1 about the fixed joint axis followed by another screw displacement of S_2 about the displaced moving joint axis. This is because a rotation of θ_1 about the fixed joint axis will displace the moving joint axis, S_2 , to a new location, and this must be accounted for the subsequent displacement.

The best scheme is to rotate the rigid body σ about the moving joint axis followed by another rotation about the fixed joint axis. In this way, the initial location of the moving joint axis can be used for derivation of the transformation matrices, and the resulting transformation matrix can be obtained by a premultiplication of the two successive screw displacements; that is,

$$A_r = A_1 A_2, \quad (2.138)$$

where A_1 and A_2 , denote the screw displacements about the initial locations of the fixed and moving joint axes, respectively.

The commutative property with regard to the change in joint variables in an open-loop chain should not be confused with the fact that finite rotations about body-fixed (or space fixed) axes do not commute. This property was first discovered and applied to the kinematic synthesis of spatial linkages by Tsai and Roth (Tsai, 1972; Tsai and Roth, 1972, 1973). Gupta (1984, 1986, 1997) subsequently provided a more rigorous proof of the property and applied it for the kinematic analysis of serial manipulators.

The proposition above can be extended to an n -link serial manipulator as shown in Fig. 2.14. Toward this goal, we first define a *reference position* for the manipulator. Although the reference position can be chosen arbitrarily, it is usually chosen at a location where the coordinates of all the joint axes can easily be identified. The reference position is sometimes called the *zero position*, and the desired position is called the *target position*. In this way, we may consider the manipulator as being displaced from the reference position

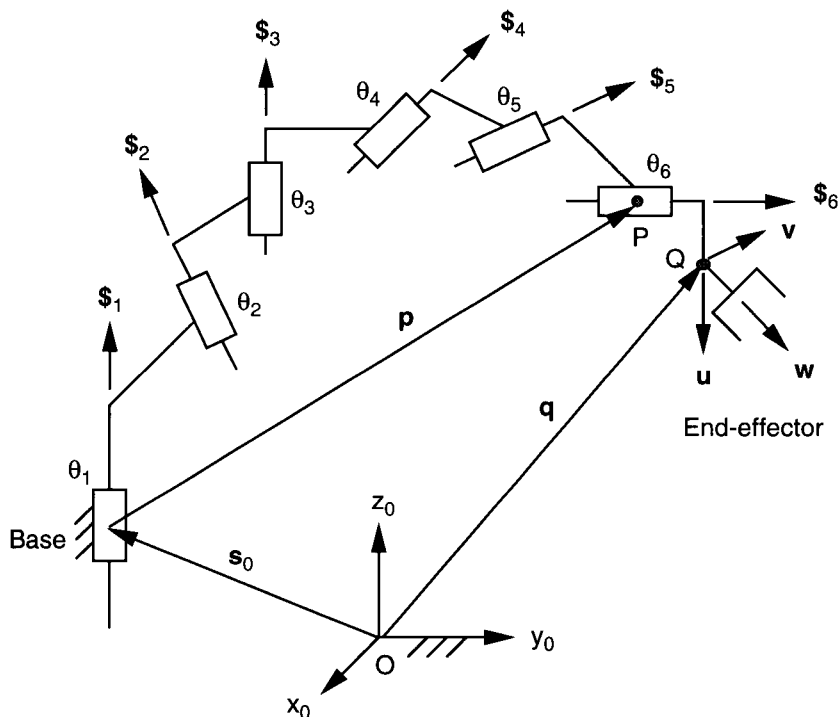


FIGURE 2.14. Screw axes of a serial manipulator.

to the target position by a series of finite screw displacements about all the joint axes.

(a) Reference Position. The reference position of a manipulator is specified in terms of the end effector location,

$$u_0, v_0, w_0, \text{ and } q_0,$$

and the direction and orientation of the joint axes:

$$s_i \text{ and } s_{oi}, \text{ for } i = 1, 2, \dots, n.$$

(b) Target Position. The target position of a manipulator is specified in terms of the desired end-effector location:

$$u, v, w, \text{ and } q.$$

(c) Loop-Closure Equation. The end-effector displacement from a reference position to a target position can be considered as the resultant of n

successive screw displacements: that is, a rotation about the n th joint axis, followed by another about the $(n - 1)$ th joint axis, and so on. Since all screw displacements take place about the joint axes at the reference position, the resulting screw displacement is obtained by premultiplying these screw displacements:

$$A_h = A_1 A_2 \cdots A_{n-1} A_n. \quad (2.139)$$

Using the method of successive screw displacements, only one fixed coordinate system and one end effector coordinate system are needed. The screw parameters used in Eq. (2.132) should not be confused with the Denavit–Hartenberg parameters. The joint variables of a screw displacement represent the actual angles of rotation and/or distances of translation needed to bring the end effector from a reference position to a target position. Specifically, for a revolute joint, θ_i is a variable and $t_i = 0$ identically, while for a prismatic joint, t_i is a variable and $\theta_i = 0$ identically.

The D-H parameters do not represent the angle of rotation or the distance of translation about a joint axis. To obtain the actual displacements, it is necessary to subtract the joint variables associated with a reference position from that of a target position. One of the advantages of using successive screw displacements is that the reference position can be chosen arbitrarily. For example, it can be chosen at the *home position* of a robot, where all the information regarding the location of the end effector and the locations of the joint axes are known.

For direct kinematics, we compute Eq. (2.139) directly by using the given joint variables. For inverse kinematics, the left-hand side of Eq. (2.139) is given and the problem is to find the joint displacements needed to bring the hand to a desired location.

2.7.3 Position Analysis of an Elbow Manipulator

Figure 2.15 shows the schematic diagram of an elbow manipulator. In this manipulator, the second joint axis intersects the first perpendicularly, the third and fourth joint axes are parallel to the second, the fifth joint axis is perpendicular to the fourth with a small offset distance a_4 , and the sixth joint axis intersects the fifth perpendicularly. We wish to solve the inverse kinematics problem of this manipulator using the method of successive screw displacements.

(a) Reference Position. First we identify a reference configuration with respect to which the displacement of the manipulator will be measured. Figure 2.16 shows such a reference configuration, where the first joint axis, S_1 ,

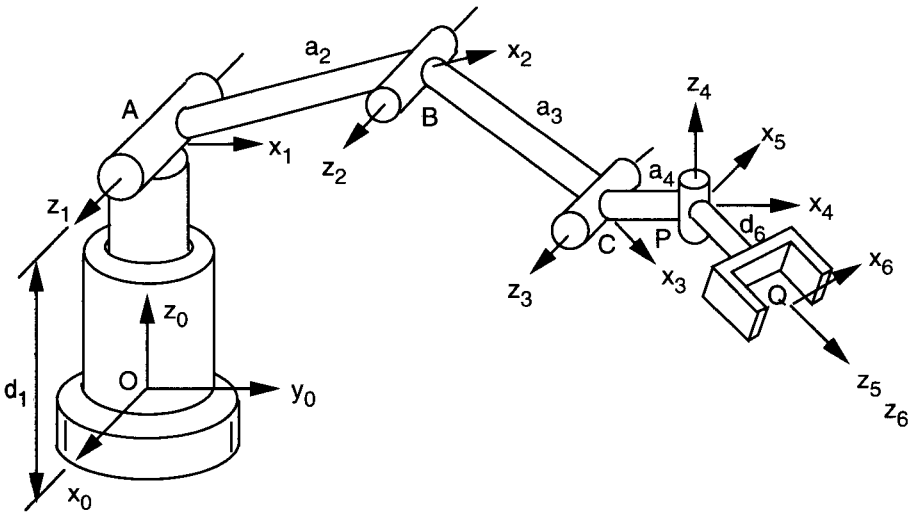


FIGURE 2.15. A 6-dof elbow manipulator.

points up vertically in the positive z -direction; the second, third, and fourth joint axes, $\$2$, $\$3$, and $\$4$, are all pointing out of the paper; the fifth joint axis, $\$5$, points in the positive z -direction; and the sixth joint axis, $\$6$, points in the positive x -direction. The hand coordinate system is located at point Q such that the w_0 -axis points in the positive x -direction and the u_0 -axis points in the

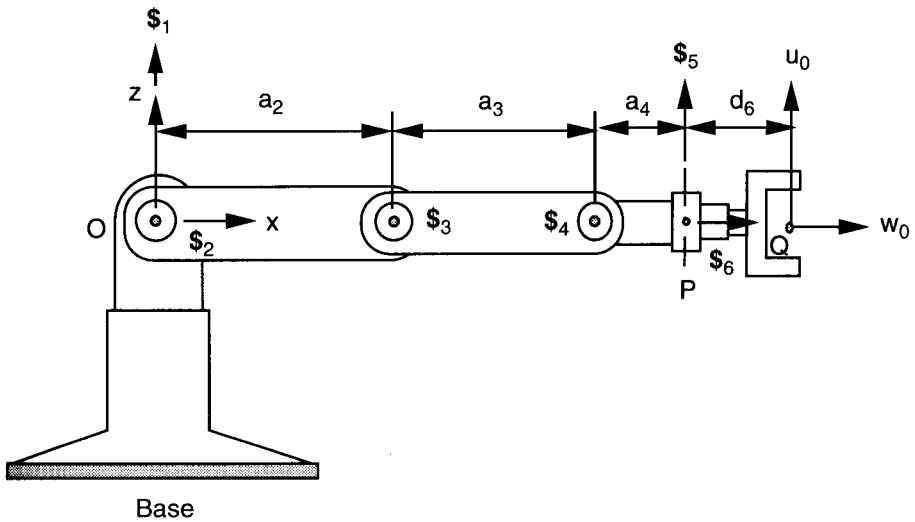


FIGURE 2.16. Reference position of the elbow manipulator.

TABLE 2.5. Screw Axis Locations of the Elbow Manipulator

Joint i	$\mathbf{s}_i$	$\mathbf{s}_{oi}$
1	(0, 0, 1)	(0, 0, 0)
2	(0, -1, 0)	(0, 0, 0)
3	(0, -1, 0)	(a_2 , 0, 0)
4	(0, -1, 0)	($a_2 + a_3$, 0, 0)
5	(0, 0, 1)	($a_2 + a_3 + a_4$, 0, 0)
6	(1, 0, 0)	(0, 0, 0)

positive z -direction. At this reference position, the locations of the screw axes with respect to the fixed reference frame are listed in Table 2.5. The reference position of the end effector is

$$\mathbf{u}_0 = [0, 0, 1]^T, \quad \mathbf{v}_0 = [0, -1, 0]^T, \quad \mathbf{w}_0 = [1, 0, 0]^T, \quad \text{and} \\ \mathbf{p}_0 = [a_2 + a_3 + a_4, 0, 0]^T.$$

(b) Target Position. Let the target position of the end effector be

$$\mathbf{u} = [u_x, u_y, u_z]^T, \quad \mathbf{v} = [v_x, v_y, v_z]^T, \quad \mathbf{w} = [w_x, w_y, w_z]^T, \quad \text{and} \\ \mathbf{p} = [p_x, p_y, p_z]^T.$$

(c) Transformation Matrices. Substituting the coordinates of the joint axes into Eq. (2.132), we obtain the screw transformation matrices:

$$A_1 = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^0A_1)^{-1} = \begin{bmatrix} c\theta_1 & s\theta_1 & 0 & 0 \\ -s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ A_2 = \begin{bmatrix} c\theta_2 & 0 & -s\theta_2 & 0 \\ 0 & 1 & 0 & 0 \\ s\theta_2 & 0 & c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \\ A_3 = \begin{bmatrix} c\theta_3 & 0 & -s\theta_3 & a_2(1 - c\theta_3) \\ 0 & 1 & 0 & 0 \\ s\theta_3 & 0 & c\theta_3 & -a_2s\theta_3 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_4 = \begin{bmatrix} c\theta_4 & 0 & -s\theta_4 & (a_2 + a_3)(1 - c\theta_4) \\ 0 & 1 & 0 & 0 \\ s\theta_4 & 0 & c\theta_4 & -(a_2 + a_3)s\theta_4 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_5 = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & (a_2 + a_3 + a_4)(1 - c\theta_5) \\ s\theta_5 & c\theta_5 & 0 & -(a_2 + a_3 + a_4)s\theta_5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

$$A_6 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta_6 & -s\theta_6 & 0 \\ 0 & s\theta_6 & c\theta_6 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The matrix products $A_2A_3A_4$ and $A_1A_2A_3A_4$ are computed as

$$A_2A_3A_4 = \begin{bmatrix} c\theta_{234} & 0 & -s\theta_{234} & a_2c\theta_2 + a_3c\theta_{23} - (a_2 + a_3)c\theta_{234} \\ 0 & 1 & 0 & 0 \\ s\theta_{234} & 0 & c\theta_{234} & a_2s\theta_2 + a_3s\theta_{23} - (a_2 + a_3)s\theta_{234} \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (2.140)$$

$$A_1A_2A_3A_4 = \begin{bmatrix} c\theta_1c\theta_{234} & -s\theta_1 & -c\theta_1s\theta_{234} & c\theta_1[a_2c\theta_2 + a_3c\theta_{23} - (a_2 + a_3)c\theta_{234}] \\ s\theta_1c\theta_{234} & c\theta_1 & -s\theta_1s\theta_{234} & s\theta_1[a_2c\theta_2 + a_3c\theta_{23} - (a_2 + a_3)c\theta_{234}] \\ s\theta_{234} & 0 & c\theta_{234} & [a_2s\theta_2 + a_3s\theta_{23} - (a_2 + a_3)s\theta_{234}] \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (2.141)$$

(d) Inverse Kinematics. The transformation of the wrist center point P is given by

$$\mathbf{p} = A_1A_2A_3A_4\mathbf{p}_0. \quad (2.142)$$

Multiplying both sides of the equation above by A_1^{-1} , we obtain

$$A_1^{-1} \begin{bmatrix} p_x \\ p_y \\ p_z \\ 1 \end{bmatrix} = A_2A_3A_4 \begin{bmatrix} a_2 + a_3 + a_4 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \quad (2.143)$$

Substituting A_1^{-1} and Eq. (2.140) into (2.143) yields

$$p_x c\theta_1 + p_y s\theta_1 = a_2 c\theta_2 + a_3 c\theta_{23} + a_4 c\theta_{234}, \quad (2.144)$$

$$-p_x s\theta_1 + p_y c\theta_1 = 0, \quad (2.145)$$

$$p_z = a_2 s\theta_2 + a_3 s\theta_{23} + a_4 s\theta_{234}. \quad (2.146)$$

From Eq. (2.145), two solutions of θ_1 are found immediately:

$$\theta_1 = \tan^{-1} \frac{p_y}{p_x}. \quad (2.147)$$

For this manipulator, the position and orientation are not decoupled. Therefore, we need to work on both simultaneously. Applying the transformation matrix to the approach vector $\mathbf{w}$ gives

$$R_1^T \mathbf{w} = R_2 R_3 R_4 R_5 \mathbf{w}_0, \quad (2.148)$$

where R_i denotes the upper left 3×3 submatrix of A_i . Expanding Eq. (2.148), we obtain

$$w_x c\theta_1 + w_y s\theta_1 = c\theta_{234} c\theta_5, \quad (2.149)$$

$$-w_x s\theta_1 + w_y c\theta_1 = s\theta_5, \quad (2.150)$$

$$w_z = s\theta_{234} c\theta_5. \quad (2.151)$$

Corresponding to each solution of θ_1 , Eq. (2.150) yields two solutions of θ_5 :

$$\theta_5 = \sin^{-1}(-w_x s\theta_1 + w_y c\theta_1). \quad (2.152)$$

That is, if $\theta_5 = \theta_5^*$ is a solution, $\theta_5 = \pi - \theta_5^*$ is also a solution. Once θ_1 and θ_5 are known, Eqs. (2.149) and (2.151) can be solved for $s\theta_{234}$ and $c\theta_{234}$. This leads to a unique solution for θ_{234} :

$$\theta_{234} = \text{Atan2} [w_z / c\theta_5, (w_x c\theta_1 + w_y s\theta_1) / c\theta_5]. \quad (2.153)$$

Next, we solve Eqs. (2.144) and (2.146) for θ_2 and θ_3 . For convenience, we rewrite Eqs. (2.144) and (2.146) as follows:

$$a_2 c\theta_2 + a_3 c\theta_{23} = k_1, \quad (2.154)$$

$$a_2 s\theta_2 + a_3 s\theta_{23} = k_2, \quad (2.155)$$

where $k_1 = p_x c\theta_1 + p_y s\theta_1 - a_4 c\theta_{234}$ and $k_2 = p_z - a_4 s\theta_{234}$. Summing the squares of Eqs. (2.154) and (2.155), we obtain

$$a_2^2 + a_3^2 + 2a_2 a_3 c\theta_3 = k_1^2 + k_2^2. \quad (2.156)$$

Hence

$$\theta_3 = \cos^{-1} \frac{k_1^2 + k_2^2 - a_2^2 - a_3^2}{2a_2 a_3}. \quad (2.157)$$

Therefore, corresponding to each solution set of θ_1 , θ_5 , and θ_{234} , there are at most two real solutions of θ_3 . Namely, if θ_3^* is a solution, $\theta_3 = -\theta_3^*$ is also a solution. Once θ_3 is known, θ_2 can be obtained by solving Eqs. (2.154) and (2.155) simultaneously for $s\theta_2$ and $c\theta_2$. This produces one solution of θ_2 . Finally, the solutions of θ_4 are obtained from the relation $\theta_4 = \theta_{234} - \theta_2 - \theta_3$.

To solve for θ_6 , we apply the transformation to the unit vector $\mathbf{u}$:

$$(R_1 R_2 R_3 R_4)^T \mathbf{u} = R_5 R_6 \mathbf{u}_0. \quad (2.158)$$

Expanding Eq. (2.158), we obtain

$$u_x c\theta_1 c\theta_{234} + u_y s\theta_1 c\theta_{234} + u_z s\theta_{234} = s\theta_5 s\theta_6, \quad (2.159)$$

$$-u_x s\theta_1 + u_y c\theta_1 = -c\theta_5 s\theta_6, \quad (2.160)$$

$$-u_x c\theta_1 s\theta_{234} - u_y s\theta_1 s\theta_{234} + u_z c\theta_{234} = c\theta_6. \quad (2.161)$$

We can solve Eqs. (2.159) and (2.160) for $s\theta_6$:

$$s\theta_6 = s\theta_5 (u_x c\theta_1 c\theta_{234} + u_y s\theta_1 c\theta_{234} + u_z s\theta_{234}) - c\theta_5 (-u_x s\theta_1 + u_y c\theta_1). \quad (2.162)$$

Equations (2.161) and (2.162) together determine a unique solution for θ_6 :

$$\theta_6 = \text{Atan2}(s\theta_6, c\theta_6). \quad (2.163)$$

We conclude that there are at most eight real inverse kinematic solutions.

2.7.4 Position Analysis of the Stanford Arm

Figure 2.17 shows a 6-dof manipulator developed at Stanford University (Scheinman, 1969). In this manipulator, the third joint is a prismatic joint (or sliding pair) while all the others are revolute. The first joint axis, $\$1$, points up vertically in the positive z -direction. The second joint axis, $\$2$, intersects the first perpendicularly at point A and points in the positive x -direction. The third joint axis intersects the second perpendicularly at point B and points

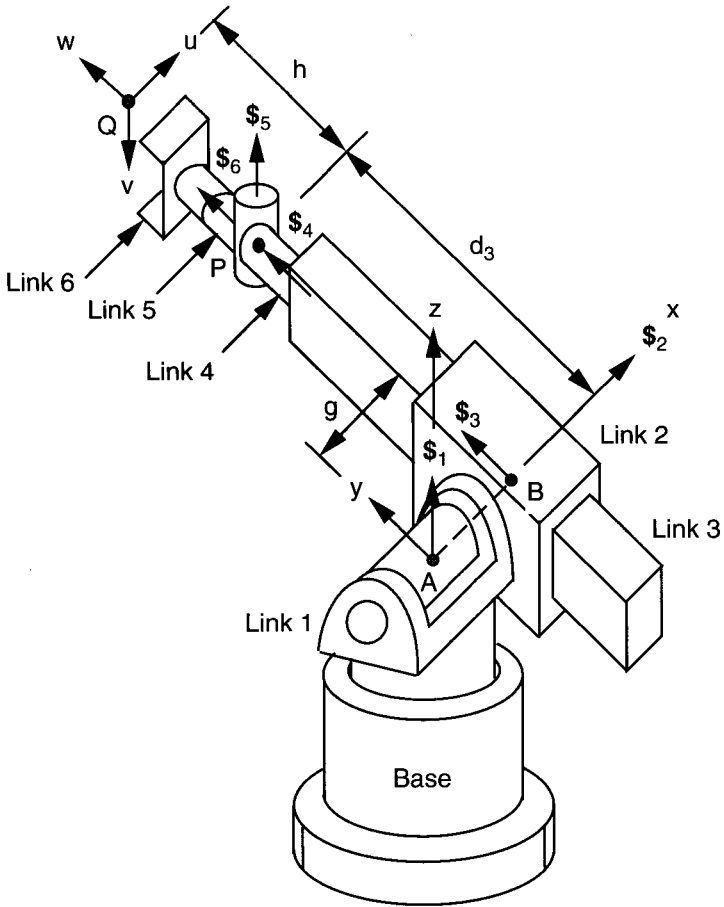


FIGURE 2.17. Screw axes associated with the Stanford manipulator.

in the positive y -direction. The fourth joint axis is in line with the third. Furthermore, the last three joint axes intersect one another perpendicularly at the wrist center point P as shown in Fig. 2.17.

(a) Reference Position. The reference position is chosen at the location where the wrist center point P is retracted back into coincident with B . At this reference position, the locations of the joint axes with respect to the fixed coordinate system are listed in Table 2.6, where g denotes the offset distance between the first and third joint axes.

The reference position of the end effector is given by

$$\mathbf{u}_0 = [1, 0, 0]^T, \quad \mathbf{v}_0 = [0, 0, -1]^T, \quad \mathbf{w}_0 = [0, 1, 0]^T, \quad \text{and} \\ \mathbf{q}_0 = [g, h, 0]^T.$$

TABLE 2.6. Screw Axis Locations of the Stanford Arm

Joint i	$\mathbf{s}_i (s_x, s_y, s_z)$	$\mathbf{s}_{oi} (s_{ox}, s_{oy}, s_{oz})$
1	(0, 0, 1)	(0, 0, 0)
2	(1, 0, 0)	(0, 0, 0)
3	(0, 1, 0)	(g , 0, 0)
4	(0, 1, 0)	(g , 0, 0)
5	(0, 0, 1)	(g , 0, 0)
6	(0, 1, 0)	(g , 0, 0)

Furthermore, the reference position of the wrist center P is given by

$$\mathbf{p}_0 = [g, 0, 0]^T.$$

(b) Target Position. Let the target position of the end effector be

$$\mathbf{u} = [u_x, u_y, u_z]^T, \quad \mathbf{v} = [v_x, v_y, v_z]^T, \quad \mathbf{w} = [w_x, w_y, w_z]^T, \quad \text{and} \\ \mathbf{q} = [q_x, q_y, q_z]^T.$$

Then the target position of the wrist center is given by

$$\mathbf{p} = \mathbf{q} - h\mathbf{w}, \quad (2.164)$$

where h is the distance between points P and Q .

In what follows, we first relate the position of the wrist center P to the first three joint variables. Then we solve the last three joint variables in terms of the end-effector orientation.

(c) Wrist Center Position. We observe that the position of the wrist center P depends only on the first three joint variables, θ_1 , θ_2 , and d_3 . Substituting the coordinates of the first three joint axes into Eq. (2.132), we obtain

$$\begin{aligned} A_1 A_2 A_3 &= \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c\theta_2 & -s\theta_2 & 0 \\ 0 & s\theta_2 & c\theta_2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c\theta_1 & -s\theta_1 c\theta_2 & s\theta_1 s\theta_2 & -d_3 s\theta_1 c\theta_2 \\ s\theta_1 & c\theta_1 c\theta_2 & -c\theta_1 s\theta_2 & d_3 c\theta_1 c\theta_2 \\ 0 & s\theta_2 & c\theta_2 & d_3 s\theta_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \end{aligned} \quad (2.165)$$

The target position of the wrist center is related to its initial reference position by

$$\mathbf{p} = A_1 A_2 A_3 \mathbf{p}_0. \quad (2.166)$$

Substituting Eq. (2.165) into (2.166) yields

$$p_x = g c \theta_1 - d_3 s \theta_1 c \theta_2, \quad (2.167)$$

$$p_y = g s \theta_1 + d_3 c \theta_1 c \theta_2, \quad (2.168)$$

$$p_z = d_3 s \theta_2. \quad (2.169)$$

We note that the distance between point A and the wrist center P as shown in Fig. 2.17 does not depend on the joint variables θ_1 and θ_2 . We can eliminate both θ_1 and θ_2 by the following operation. Summing the squares of Eqs. (2.167), (2.168), and (2.169) yields

$$p_x^2 + p_y^2 + p_z^2 = g^2 + d_3^2. \quad (2.170)$$

Hence

$$d_3 = \pm \sqrt{p_x^2 + p_y^2 + p_z^2 - g^2}. \quad (2.171)$$

Equation (2.171) yields (1) two real roots if $p_x^2 + p_y^2 + p_z^2 - g^2 > 0$, (2) one double root if $p_x^2 + p_y^2 + p_z^2 - g^2 = 0$, and (3) no real roots if $p_x^2 + p_y^2 + p_z^2 - g^2 < 0$. Although there may be two real values of d_3 , the negative d_3 cannot be realized physically because of the mechanical structure limitation of the robot. When Eq. (2.171) yields a double root, $d_3 = 0$, the wrist center is retracted to point B and the manipulator is said to be in a singular configuration. When Eq. (2.171) yields no real roots, the target point is not reachable. Furthermore, let $d_{\max}$ be the maximum extension of the prismatic joint. Then a target point is not reachable if $d_3 > d_{\max}$, even though Eq. (2.171) may yield two real roots.

Once d_3 is known, we can solve Eq. (2.169) for θ_2 as

$$\theta_2 = \sin^{-1} \frac{p_z}{d_3}. \quad (2.172)$$

Hence, corresponding to each d_3 , Eq. (2.172) yields at most two real solutions of θ_2 . In general, if $\theta_2 = \theta_2^*$ is a solution, $\theta_2 = \pi - \theta_2^*$ is also a solution. We note that $p_x^2 + p_y^2 - g^2 = 0$ represents a cylinder of radius g with the z -axis as its longitudinal axis. It can be shown that Eq. (2.172) yields (1) two real roots if the target point falls outside the cylinder ($p_x^2 + p_y^2 - g^2 > 0$), (2) one

double root if the target point falls on the cylinder ($p_x^2 + p_y^2 - g^2 = 0$), and (3) no real roots if the target point falls inside the cylinder ($p_x^2 + p_y^2 - g^2 < 0$). When the target point falls outside the cylinder, Eq. (2.171) also yields two real roots.

Once d_3 and θ_2 are known, we can solve Eqs. (2.167) and (2.168) simultaneously for $s\theta_1$ and $c\theta_1$:

$$c\theta_1 = \frac{gp_x + d_3 p_y c\theta_2}{g^2 + d_3^2 c^2 \theta_2}, \quad (2.173)$$

$$s\theta_1 = \frac{gp_y - d_3 p_x c\theta_2}{g^2 + d_3^2 c^2 \theta_2}. \quad (2.174)$$

Therefore, corresponding to each solution set of θ_2 and d_3 , Eqs. (2.173) and (2.174) yield one solution of θ_1 :

$$\theta_1 = \text{Atan2}(s\theta_1, c\theta_1). \quad (2.175)$$

We conclude that corresponding to each given end-effector location, there are at most four upper arm configurations.

(d) End-Effector Orientation. We now proceed to find the last three joint angles. The end-effector orientation depends only on the rotational part of the transformation matrices A_i . Substituting the coordinates of the last three joint axes into Eq. (2.132), we obtain

$$\begin{aligned} R_4 R_5 R_6 &= \begin{bmatrix} c\theta_4 & 0 & s\theta_4 \\ 0 & 1 & 0 \\ -s\theta_4 & 0 & c\theta_4 \end{bmatrix} \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 \\ s\theta_5 & c\theta_5 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta_6 & 0 & s\theta_6 \\ 0 & 1 & 0 \\ -s\theta_6 & 0 & c\theta_6 \end{bmatrix} \\ &= \begin{bmatrix} c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6 & -c\theta_4 s\theta_5 & c\theta_4 c\theta_5 s\theta_6 + s\theta_4 c\theta_6 \\ s\theta_5 c\theta_6 & c\theta_5 & s\theta_5 s\theta_6 \\ -s\theta_4 c\theta_5 c\theta_6 - c\theta_4 s\theta_6 & s\theta_4 s\theta_5 & -s\theta_4 c\theta_5 s\theta_6 + c\theta_4 c\theta_6 \end{bmatrix}, \end{aligned} \quad (2.176)$$

where R_i denotes the upper left 3×3 submatrix of A_i .

The transformation of the approach vector $\mathbf{w}$ is

$$\mathbf{w} = R_1 R_2 R_3 R_4 R_5 R_6 \mathbf{w}_0. \quad (2.177)$$

Multiplying both sides of Eq. (2.177) by $(R_1 R_2 R_3)^{-1}$, we obtain

$$R_3^T R_2^T R_1^T \mathbf{w} = R_4 R_5 R_6 \mathbf{w}_0. \quad (2.178)$$

The quantities on the left-hand side of Eq. (2.178) are known. For convenience, we define ${}^3\mathbf{w} \equiv R_3^T R_2^T R_1^T \mathbf{w}$. Expanding Eq. (2.178), we obtain

$$-c\theta_4 s\theta_5 = {}^3w_x, \quad (2.179)$$

$$c\theta_5 = {}^3w_y, \quad (2.180)$$

$$s\theta_4 s\theta_5 = {}^3w_z, \quad (2.181)$$

where

$${}^3w_x = w_x c\theta_1 + w_y s\theta_1,$$

$${}^3w_y = (-w_x s\theta_1 + w_y c\theta_1) c\theta_2 + w_z s\theta_2,$$

$${}^3w_z = (w_x s\theta_1 - w_y c\theta_1) s\theta_2 + w_z c\theta_2.$$

Note that the approach vector $\mathbf{w}$ does not depend on the sixth joint angle, θ_6 . Solving Eq. (2.180) for θ_5 yields

$$\theta_5 = \cos^{-1}({}^3w_y). \quad (2.182)$$

Equation (2.182) yields two real roots of θ_5 if $|{}^3w_y| < 1$. When $|{}^3w_y| = 1$, Eq. (2.182) yields $\theta_5 = 0$ or π . For this case the sixth joint axis is in line with the fourth, and the wrist is said to be at a singular configuration. The condition $|{}^3w_y| > 1$ cannot occur physically. Assuming that $s\theta_5 \neq 0$, we solve θ_4 from Eqs. (2.179) and (2.181):

$$\theta_4 = \text{Atan2}({}^3w_z/s\theta_5, -{}^3w_x/s\theta_5). \quad (2.183)$$

Finally, we solve θ_6 by applying the transformation to the unit vector $\mathbf{u}$.

$$\mathbf{u} = R_1 R_2 R_3 R_4 R_5 R_6 \mathbf{u}_0. \quad (2.184)$$

Multiplying both sides of Eq. (2.184) by $(R_1 R_2 R_3)^{-1}$, we obtain

$$R_3^T R_2^T R_1^T \mathbf{u} = R_4 R_5 R_6 \mathbf{u}_0. \quad (2.185)$$

For convenience, we define ${}^3\mathbf{u} \equiv R_3^T R_2^T R_1^T \mathbf{u}$. Expanding Eq. (2.185) yields

$${}^3u_x = c\theta_4 c\theta_5 c\theta_6 - s\theta_4 s\theta_6, \quad (2.186)$$

$${}^3u_y = s\theta_5 c\theta_6, \quad (2.187)$$

$${}^3u_z = -s\theta_4 c\theta_5 c\theta_6 - c\theta_4 s\theta_6, \quad (2.188)$$

where

$$\begin{aligned} {}^3u_x &= u_x c\theta_1 + u_y s\theta_1, \\ {}^3u_y &= (-u_x s\theta_1 + u_y c\theta_1) c\theta_2 + u_z s\theta_2, \\ {}^3u_z &= (u_x s\theta_1 - u_y c\theta_1) s\theta_2 + u_z c\theta_2. \end{aligned}$$

Multiplying Eq. (2.186) by $s\theta_4$ and (2.188) by $c\theta_4$ and summing produces

$$s\theta_4 {}^3u_x + c\theta_4 {}^3u_z = -s\theta_6. \quad (2.189)$$

From Eqs. (2.187) and (2.189), we obtain a unique solution for θ_6 :

$$\theta_6 = \text{Atan2}(-s\theta_4 {}^3u_x - c\theta_4 {}^3u_z, {}^3u_y/s\theta_5). \quad (2.190)$$

We conclude that corresponding to a given end-effector location, there are up to eight inverse kinematics solutions, but only four are physically realizable.

2.8 SUMMARY

In this chapter, two methods for solving the inverse kinematics of serial manipulators have been presented. First, the Denavit–Hartenberg parameters were defined. Then the D-H transformation matrices were developed for derivation of the kinematic equations. The D-H method was demonstrated by a planar 3-dof manipulator, the Scorbot robot, and the Fanuc S-900W robot. In addition, Tsai and Morgan's solution for the cases of three intersecting joint axes and three parallel joint axes was presented. Second, the concept of successive screw displacements and its application to the inverse kinematics of serial manipulators were described. It is shown that the inverse kinematics can be solved in a straightforward manner by applying the method of successive screw displacements. The inverse kinematics of an elbow manipulator and the Stanford manipulator were solved to illustrate the concept.

REFERENCES

- Albala, H., 1982, "Displacement Analysis of the General N -Bar, Single-Loop, Spatial Linkages," *ASME J. Mech. Des.*, Vol. 104, No. 2, pp. 504–525.
- Albala, H. and Angeles, J., 1979, "Numerical Solution to the Input–Output Displacement Equation of the General $7R$ Spatial Mechanism," *Proc. 5th World Congress on Theory of Machines and Mechanisms*, pp. 1008–1011.
- Bottema, O. and Roth, B., 1979, *Theoretical Kinematics*, North Holland, New York.

- Chase, M. A., 1963, "Vector Analysis of Linkages," *ASME J. Eng. Ind.*, Vol. 85, pp. 289–297.
- Denavit, J. and Hartenberg, R. S., 1955, "A Kinematic Notation for Lower Pair Mechanisms Based on Matrices," *ASME J. Appl. Mech.*, Vol. 77, pp. 215–221.
- Duffy, J., 1980, *Analysis of Mechanisms and Robot Manipulators*, Wiley, New York.
- Duffy, J. and Crane, C., 1980, "A Displacement Analysis of the General Spatial 7R Mechanism," *Mech. Mach. Theory*, Vol. 15, pp. 153–169.
- Duffy, J. and Rooney, J., 1975, "A Foundation for a Unified Theory of Analysis of Spatial Mechanisms," *ASME J. Eng. Ind.*, Vol. 97, No. 4, pp. 1159–1164.
- Freudenstein, F., 1973, "Kinematics: Past, Present, and Future," *Mech. Mach. Theory*, Vol. 8, No. 2, pp. 151–161.
- Gupta, K. C., 1984, "A Note on Position Analysis of Manipulators," *Mech. Mach. Theory*, Vol. 19, pp. 5–8.
- Gupta, K. C., 1986, "Kinematic Analysis of Manipulators Using Zero Reference Position Description," *Int. J. Robot. Res.*, Vol. 5, No. 2, pp. 5–13.
- Gupta, K. C., 1997, *Mechanics and Control of Robots*, Springer-Verlag, New York.
- Kohli, D. and Soni, A. H., 1975, "Kinematic Analysis of Spatial Mechanisms via Successive Screw Displacements," *ASME J. Eng. Ind.*, Vol. 97, No. 2, pp. 739–747.
- Lee, H. Y. and Liang, C. G., 1988a, "A New Vector Theory for the Analysis of Spatial Mechanisms," *Mech. Mach. Theory*, Vol. 23, No. 3, pp. 209–217.
- Lee, H. Y. and Liang, C. G., 1988b, "Displacement Analysis of the General Spatial 7-Link 7R Mechanism," *Mech. Mach. Theory*, Vol. 23, No. 3, pp. 219–226.
- Pennock, G. R. and Yang, A. T., 1985, "Application of Dual-Number Matrices to the Inverse Kinematics Problem of Robot Manipulators," *ASME J. Mech. Transm. Autom. Des.*, Vol. 107, No. 2, pp. 201–208.
- Pieper, D., 1968, "The Kinematics of Manipulators Under Computer Control," Ph.D. dissertation, Department of Mechanical Engineering, Stanford University, Stanford, CA.
- Pieper, D. L. and Roth, B., 1969, "The Kinematics of Manipulators Under Computer Control," *Proc. 2nd International Congress for the Theory of Machines and Mechanisms*, Zakopane, Poland, Vol. 2, pp. 159–168.
- Primrose, E. J. F., 1986, "On the Input–Output Equation of the General 7R-Mechanism," *Mech. Mach. Theory*, Vol. 21, No. 6, pp. 509–510.
- Raghavan, M. and Roth, B., 1989, "Kinematic Analysis of the 6R Manipulator of General Geometry," *Proc. 5th International Symposium on Robotics Research*, edited by H. Miura and S. Arimoto, MIT Press, Cambridge, MA; preprint 1989, final 1990.
- Raghavan, M. and Roth, B., 1990, "A General Solution for the Inverse Kinematics of All Series Chains," *Proc. 8th CISM-IFTOMM Symposium on Robots and Manipulators (ROMANSY-90)*, Cracow, Poland.

- Roth, B., Rastegar, J., and Scheinman, V., 1973, "On the Design of Computer Controlled Manipulators; On the Theory and Practice of Robots and Manipulators," *Proc. First CISM-IFTOMM Symposium*, Vol. 1, pp. 93–113.
- Scheinman, V., 1969, "Design of Computer Controlled Manipulators," Report AIM 92, Stanford Artificial Intelligence Laboratory, Stanford, CA.
- Tsai, L. W., 1972, "Design of Open-Loop Chains for Rigid Body Guidance," Ph.D. dissertation, Department of Mechanical Engineering, Stanford University, Stanford, CA.
- Tsai, L. W. and Morgan, A., 1985, "Solving the Kinematics of the Most General Six- and Five-Degree-of-Freedom Manipulators by Continuation Methods," *ASME J. Mech. Transm. Autom. Des.*, Vol. 107, June, pp. 189–200.
- Tsai, L. W. and Roth, B., 1972, "Design of Dyads with Helical, Cylindrical, Spherical, Revolute, and Prismatic Joints," *Mech. Mach. Theory*, Vol. 7, pp. 85–102.
- Tsai, L. W. and Roth, B., 1973, "Incompletely Specified Displacement: Geometry and Spatial Linkage Synthesis," *ASME J. Eng. Ind.*, Vol. 95, Ser. B, No. 2, pp. 603–611.
- Uicker, J. J., Denavit, J., and Hartenberg, R. S., 1964, "An Iterative Method for Displacement Analysis of Spatial Mechanisms," *ASME J. Appl. Mech.*, Vol. 86, Ser. E, No. 2, pp. 309–314.
- Yang, A. T., 1969, "Displacement Analysis of Spatial Five-Link Mechanisms Using (3×3) Matrices with Dual-Number Elements," *ASME J. Eng. Ind.*, Vol. 9, No. 1, pp. 152–157.
- Yang, A. T. and Freudenstein, F., 1964, "Application of Dual Number Quaternion Algebra to the Analysis of Spatial Mechanisms," *ASME J. Appl. Mech.*, Vol. 86, pp. 300–308.
- Yuan, M. S. C. and Freudenstein, F., 1971, "Kinematic Analysis of Spatial Mechanisms by Means of Screw Coordinates," *ASME J. Eng. Ind.*, Vol. 93, No. 1, pp. 61–73.

EXERCISES

- Figure 2.18 shows an alternative method of coordinate frame assignment for the SCARA arm. Establish the D-H transformation matrices. Compare these transformation matrices with those given in Eqs. (2.9) through (2.12). Are there other methods of assigning the coordinate frames?
- Figure 2.19 shows a spatial 3-dof, 3R manipulator where the second joint axis is perpendicular to the first while the third joint axis is parallel to the second. Establish the D-H and overall transformation matrices ${}^{i-1}A_i$ for $i = 1, 2, \dots, 4$ and 0A_4 .
- For the elbow manipulator shown in Fig. 2.15, establish the D-H and overall transformation matrices ${}^{i-1}A_i$ for $i = 1, 2, \dots, 6$ and 0A_6 .

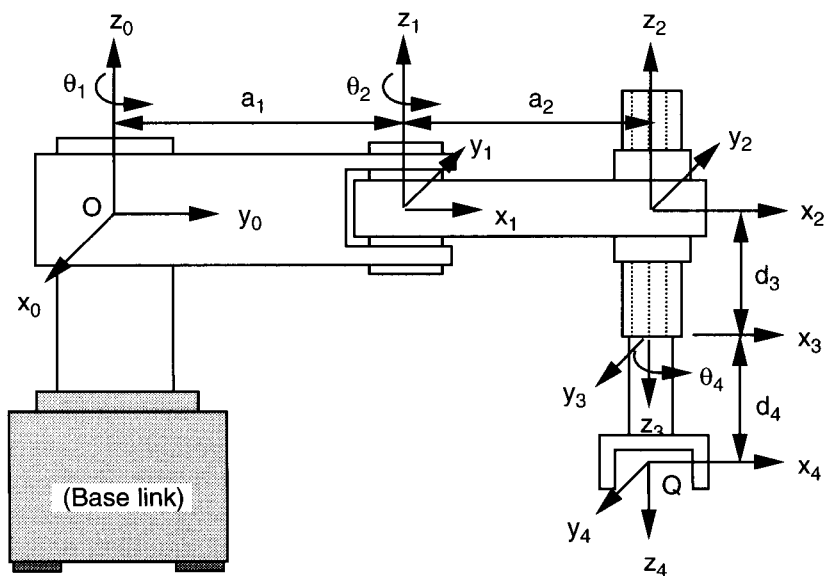


FIGURE 2.18. Alternative method of frame assignment for the SCARA arm.

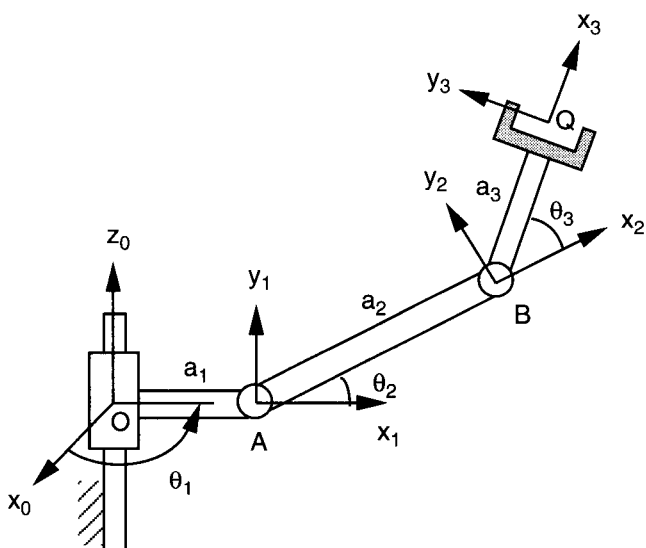


FIGURE 2.19. Spatial 3-dof, 3R manipulator.

4. Find the joint angles needed to bring the end effector of the manipulator shown in Fig. 2.19 to a given position $\mathbf{q}$ by the D-H method. Corresponding to a given position of $\mathbf{q}$, how many number of possible arm configurations exist?
5. Solve the inverse kinematics of the 4-dof SCARA arm shown in Fig. 2.4 using the D-H method. Discuss how the orientation of the end effector can be specified.
6. For the elbow-type manipulator shown in Fig. 2.15, solve the joint angles in terms of a given end-effector location, $\mathbf{q}$ and $\mathbf{u}, \mathbf{v}, \mathbf{w}$. How many possible solutions exist?
7. Figure 2.20 shows a schematic diagram of the Stanford arm. Find the joint angles needed to bring the end effector to a given position and orien-

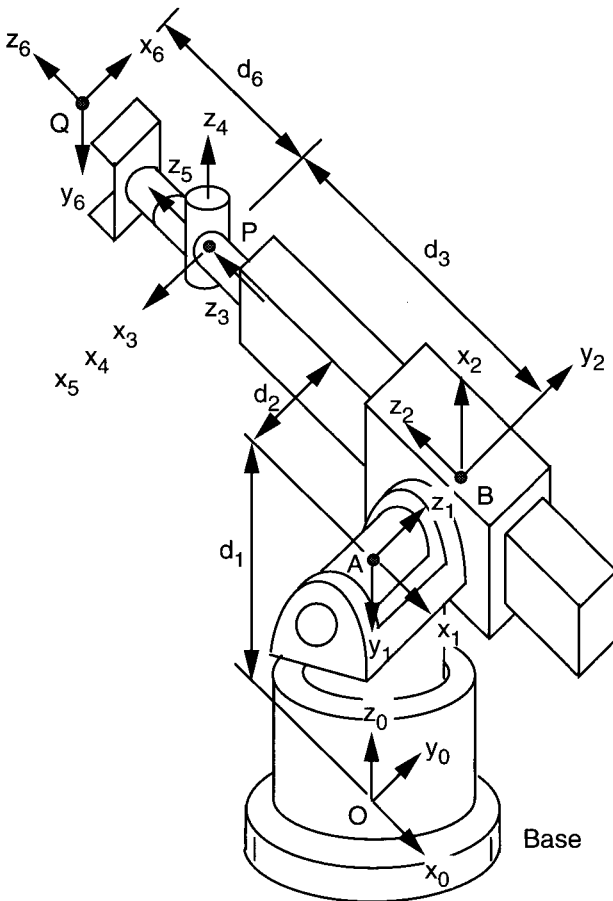


FIGURE 2.20. D-H coordinate systems associated with the Stanford manipulator.

- tation, $\mathbf{q}$ and $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$. Under what conditions will the manipulator possess no real solutions?
8. Solve the inverse kinematics of the 3-dof manipulator shown in Fig. 2.19 using the vector-loop approach.
 9. Solve the inverse kinematics of the 4-dof SCARA arm shown in Fig. 2.4 using the vector-loop approach.
 10. For the 3-dof manipulator shown in Fig. 2.19, find the joint angles needed to bring the end effector to a given position $\mathbf{q}$ using the method of successive screw displacements.
 11. Figure 2.21 shows a reference position of the Scorbot robot. At this reference position, the first joint axis points up vertically along the positive z -direction; the second, third, and fourth joint axes point out of the paper, the fifth joint axis points to the right and intersects the fourth at P . The wrist center is located at Q . Solve the inverse kinematics problem using the method of successive screw displacements.

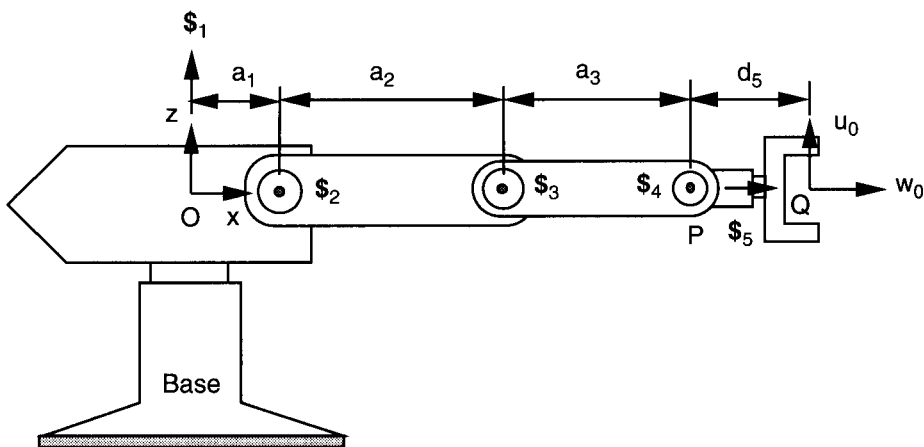


FIGURE 2.21. Reference position of the Scorbot robot.

12. Figure 2.22 shows a reference position of the Fanuc S-900W robot. Solve the inverse kinematics using the method of successive screw displacements.

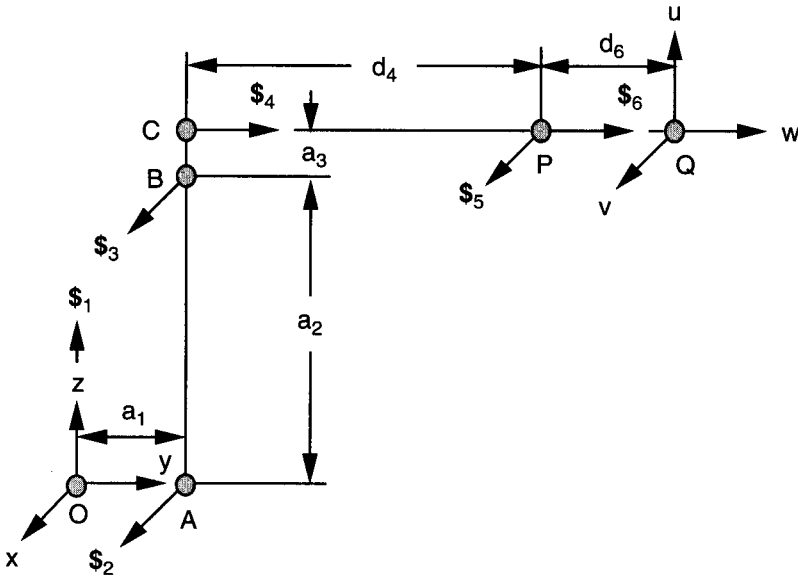


FIGURE 2.22. Reference position of the Fanuc S-900W robot.